

# Revisiting Fundamentals of the European Gas Market: The Role of Supply Substitution

## Abstract

We study the European natural gas market through a Bayesian Structural Vector Autoregression that distinguishes pipeline gas from liquefied natural gas (LNG), with contemporaneous elasticities identified through sign and elasticity restrictions. Pipeline gas is highly price-inelastic and propagates supply disruptions as transient spikes, whereas LNG acts as the flexible balancing margin with more persistent price effects. Three counterfactual exercises quantify the contributions of supply substitution and demand adjustment during the 2022 energy crisis. Preventing European LNG imports from expanding during the crisis raises the real price by roughly 120% relative to its observed path. At the opposite extreme, allowing LNG imports to fully absorb the contemporaneous pipeline-and-domestic shortfall lowers prices by about 87%. Eliminating the autonomous demand-reduction signal identified by the model over the formal EU policy window raises prices by approximately 28%. The realised path lies closer to the LNG-frozen upper bound than to the LNG-unconstrained lower bound, consistent with Europe’s limited regasification capacity binding during the crisis.

## 1 Introduction

The 2022 European energy crisis brought the role of supply composition in natural gas markets into unusually sharp relief. Within a few months, deliveries through the Russian pipeline network collapsed from more than 40% of total European net imports to a residual share, and liquefied natural gas (LNG) imports rose by over 60% to partly absorb the resulting shortfall. The composition of European gas supply shifted more between February 2022 and December 2023 than over the preceding decade. The episode raised three questions for which the existing empirical literature is poorly equipped: how much of the price spike was attributable to the supply disruption itself, and how much to the inability of the substitute margin to scale up fast enough; what would prices have been under alternative LNG paths; and how much of the realised stabilisation came from supply substitution rather than from autonomous demand reduction.

The macroeconomic stakes of these questions are first-order. Natural gas is a substantial component of European headline inflation and was the dominant proximate driver of the surge in euro-area HICP from 5.1% in January 2022 to a peak of 10.6% in October 2022. The European Central Bank’s response, the fastest tightening cycle in its history, raised the deposit-facility rate from  $-0.5\%$  to  $4.0\%$  between July 2022 and September 2023, with energy-price persistence cited repeatedly as a central concern in policy communications (Adolfson et al., 2024). Whether a given gas-market disturbance generates transient or persistent inflationary pressure depends in turn on the structural composition of the supply disruption: pipeline disruptions transmit, on the evidence we report below, as relatively short-lived price spikes that the SVAR identifies as largely dissipating within five months, whereas LNG-supply shocks shift the medium-run market-clearing condition and propagate more persistently. Distinguishing the two transmission channels is therefore not only an

academic refinement but a precondition for the calibration of monetary-policy responses to gas-market disruptions, of which the ongoing 2026 Strait of Hormuz episode is the most recent illustration.

Natural gas accounts for roughly 25% of total energy consumption in the European Union and is predominantly imported. Two physically and contractually distinct channels deliver these imports. Pipeline gas is governed by long-term, route-specific contracts with take-or-pay and destination clauses, and is constrained by fixed transport infrastructure with limited spare capacity. LNG, by contrast, is increasingly traded on shorter-duration spot or hub-indexed contracts, with cargoes that can be redirected across regions in response to price differentials (Sharples, 2023; Ritz, 2019; Molnar, 2022). The two channels differ markedly in short-run elasticity, adjustment cost and contractual rigidity. Until 2018, pipeline gas accounted for over 85% of European net imports; by 2024 its share had fallen below 60%. This structural shift in supply composition, together with the abrupt and asymmetric nature of the 2022 disruption, makes the explicit identification of supply by source not a refinement of existing models but a precondition for sensible inference about gas-price formation.

Most empirical work on natural gas adapts the Structural Vector Autoregression (SVAR) framework developed for the crude-oil market by Kilian (2009); Kilian and Murphy (2014); Baumeister and Hamilton (2019), in which supply is treated as a single aggregate. In oil this is a defensible approximation; in gas, where the two supply channels respond differently to the same shocks and where the share of each has been shifting structurally, it is not. Aggregation conceals the compositional adjustment that drove the 2022 dynamics and obscures the heterogeneous transmission of supply shocks across horizons. Recent contributions emphasise that ignoring this margin systematically understates the role of supply-side forces (Moll et al., 2023; Hamilton, 2023), and that the 2022 crisis was as much a substitution episode as a level-shock episode (Güntner et al., 2024). The contribution of this paper is to model the two supply channels explicitly within a Bayesian SVAR framework that delivers economically meaningful identification of their contemporaneous elasticities.

The identification strategy follows the Bayesian sign- and elasticity-restriction approach of Baumeister and Hamilton (2015, 2019), in which informative priors over the contemporaneous structural matrix combine with the likelihood to deliver set-identified parameters whose posteriors remain asymptotically responsive to prior information. The approach is well suited to natural-gas inference for two reasons. First, short-run supply and demand elasticities in gas markets are notoriously difficult to estimate from the data alone, since supply and demand shocks generate observationally similar correlations between price and quantity. Second, the alternative of purely agnostic sign restrictions provides too little discipline in a market where the relevant elasticities are small in absolute value. A related Bayesian framework has been applied to the United States natural gas market by Rubaszek et al. (2021) and Farag (2024); this paper is, to our knowledge, the first application to the European market with separated pipeline and LNG supply channels. The dataset is constructed from JODI-Gas reports covering 22 EU member states between January 2010 and the most recent month of complete coverage, complemented by Dutch TTF prices from the World Bank Commodity Price Database and the euro-area industrial production index from Eurostat.

Three findings stand out from the posterior estimates and the structural analysis. First, the two supply channels display sharply different short-run elasticities. The posterior median for pipeline-and-domestic supply is  $\gamma_{qp} \approx 0.08$ , near-rigid, while LNG supply is substantially more responsive at  $\beta_{qp} \approx 1.6$ , consistent with the literature documenting LNG as the flexible balancing margin in Europe. Second, the impulse-response analysis shows that pipeline supply shocks transmit to prices as transient spikes that largely dissipate within five months, whereas LNG supply shocks transmit more persistently by shifting the medium-run market-clearing condition. Third, the forecast-error variance decomposition assigns roughly comparable short-horizon shares to supply and demand shocks—in contrast to much of the oil-market literature, in which demand-driven shocks dominate—and identifies storage-related precautionary demand as a significant amplification channel during episodes of supply stress.

We exploit the identified structural model to construct three counterfactual scenarios for the 2022 energy crisis. Holding LNG imports frozen at their February 2022 level for the duration of the crisis window produces a median price gap of approximately +0.8 log-points by the end of the sample: prices would have been roughly 2.2 times the realised level. Allowing LNG to fully absorb the contemporaneous pipeline-and-domestic shortfall, abstracting from Europe’s regasification bottleneck, produces a gap of  $-2.0$  log-points (prices would have been about 13.5% of the realised level). Eliminating the autonomous demand-reduction signal identified by the SVAR over the formal EU policy window of Council Regulation 2022/1369 produces a gap of +0.25 log-points. The realised price path lies closer to the LNG-frozen upper envelope than to the LNG-unconstrained lower envelope, an asymmetry consistent with the regasification bottleneck having bound during the crisis: cargoes were globally redirectable, but Europe’s capacity to receive and distribute them was operationally constrained.

The remainder of the paper is organised as follows. Section 2 reviews the related literature on energy-market SVARs and on the contractual and infrastructural differences between pipeline gas and LNG. Section 3 presents the econometric methodology, including the prior, likelihood and posterior of the Baumeister–Hamilton framework. Section 4 describes the data, the empirical specification of the contemporaneous structure, and the priors imposed on the elasticities. Section 5 reports the posterior estimates, impulse responses, forecast-error variance decompositions, historical decomposition, and the three counterfactual scenarios, together with a synthesis of their joint reading. Section 6 concludes.

## 2 Literature Review

This chapter relates the present study to three strands of literature. First, it reviews the use of Structural Vector Autoregressive (SVAR) models in the crude oil market, which provides the conceptual and methodological foundation for most empirical work on energy price shocks. Second, it surveys the growing literature applying SVARs to natural gas markets, with primary emphasis on Europe, complemented by evidence from the United States and, to a lesser extent, Asia. Finally, it reviews the literature on contractual and infrastructural differences between pipeline gas and liquefied natural gas (LNG), which motivates the explicit separation of supply sources in the empirical model.

The modern empirical analysis of energy price shocks originates in the crude oil literature. Seminal work by Hamilton (1996) documented a strong negative relationship between oil price increases and U.S. macroeconomic activity, giving rise to an extensive literature on the identification and transmission of oil price shocks. A key methodological breakthrough was introduced by Kilian (2009), who proposed a Structural VAR framework distinguishing between oil supply shocks, global aggregate demand shocks, and oil-specific demand shocks related to precautionary or speculative motives. This approach emphasized that oil price increases can have markedly different macroeconomic effects depending on their underlying source. Subsequent contributions by Kilian and Murphy (2012) and Kilian and Murphy (2014) highlighted limitations of purely agnostic sign restrictions and argued for incorporating economically meaningful restrictions on short-run elasticities and inventory behavior. Building on these insights, Baumeister and Hamilton (2015) and Baumeister and Hamilton (2019) developed a Bayesian SVAR framework under set identification, showing that informative priors on structural elasticities can discipline the identification problem without imposing exact zero restrictions. Applied to the global oil market, this methodology revealed that demand-driven shocks account for the bulk of oil price fluctuations, while supply shocks play a more limited role.

Three lessons from this literature are directly relevant for natural-gas inference: separating supply- and demand-driven shocks is essential for understanding price dynamics; short-run elasticities of energy supply and demand are typically very small, so that even moderate shocks generate large price movements; and credible identification often requires informative priors, sign restrictions or auxiliary observables such as inventories.

In contrast to crude oil, European natural gas markets have historically been regionally fragmented owing to transportation constraints and infrastructure specificity. European supply has long relied on pipeline imports under rigid long-term contracts, with LNG playing a limited role until recently, and early empirical work focused on the linkage between gas and oil prices under oil-indexed contracts; Szafranek and Rubaszek (2024) document that this relationship weakened markedly from the mid-2000s onward, reflecting market liberalisation and the transition to hub-based pricing. Recent SVAR analyses of the European market identify supply shocks, demand shocks related to economic activity or weather, and storage-driven precautionary demand, and typically find relatively inelastic short-run demand together with constrained supply responses. The 2022 energy crisis has sharpened the role of supply-side and substitution mechanisms within this framework. Güntner et al. (2024) show that the collapse in Russian pipeline deliveries represented a dominant negative supply shock partly offset by LNG inflows, with precautionary storage demand amplifying the price response. A central implication, emphasised also by Moll et al. (2023) and Hamilton (2023), is that aggregating pipeline gas and LNG into a single supply measure conceals large compositional adjustments and understates the role of supply-side shocks, motivating models that distinguish between supply sources explicitly.

The U.S. natural gas market provides a useful comparator. It is largely self-sufficient, with price formation dominated by domestic market-based production, and SVAR analyses there have focused on disentangling domestic supply shocks—in particular those associated with the shale-gas revolution—from demand shocks driven by weather, economic activity and

storage behaviour. Rubaszek et al. (2021) estimate short-run demand elasticities larger in absolute value than those typically found for oil, suggesting that fuel-switching in electricity generation and industrial use renders U.S. gas demand more price-responsive than in Europe; they also document a non-negligible role for storage-related demand shocks. Farag (2024) highlight the growing international dimension of the U.S. market following the expansion of LNG export capacity: as the United States has become a major LNG exporter, domestic prices have become increasingly exposed to global demand conditions, though the magnitude of cross-regional transmission remains smaller than in more import-dependent regions.

A large body of research shows that pipeline gas and LNG represent fundamentally different supply margins. Pipeline gas is typically governed by long-term, route-specific contracts with take-or-pay and destination clauses, which imply low short-run price elasticity and limited flexibility in response to shocks Molnar (2022). LNG supply, by contrast, has become increasingly flexible as contract durations shorten, destination restrictions weaken, and spot market trading expands, allowing cargoes to be redirected across regions in response to price differentials Sharples (2023). This flexibility gives LNG a strategic role beyond its volume share, as it provides a credible outside option that constrains the market power of pipeline exporters Ritz (2019). Evidence from the 2022 European energy crisis illustrates these mechanisms clearly: rising LNG imports partially offset the collapse in Russian pipeline flows, though only at exceptionally high spot prices, underscoring both the substitutability and the distinct pricing dynamics of the two supply channels.

Taken together, this literature implies that pipeline gas and LNG differ markedly in contractual rigidity, cost structures, elasticity, and adjustment speed. Ignoring these differences risks mischaracterizing both the sources of natural gas price fluctuations and their macroeconomic consequences. This insight provides a central motivation for the empirical strategy adopted in this chapter, which explicitly distinguishes between pipeline and LNG supplies within a Bayesian SVAR framework.

### 3 Methodology

The model can be rewritten in matrix form as a typical structural VAR (SVAR).

$$\mathbf{A}y_t = \mathbf{B}x_{t-1} + u_t \quad (1)$$

$$u_t \sim N(0_{n \times 1}, \mathbf{D}) \quad (2)$$

where  $y_t = (y_{1t}, y_{2t}, \dots, y_{nt})'$  is the  $n \times 1$  vector of endogenous variables,  $\mathbf{A}$  is the  $n \times n$  matrix that contains all the contemporaneous elasticities across the endogenous variables,  $x_{t-1} = (y_{t-1}, \dots, y_{t-m})'$  is the  $k \times 1$  vector of lagged regressor,  $\mathbf{B}$  is the  $n \times k$  autoregressive matrix of coefficients and  $\mathbf{D} = \text{diag}(d_{11}, \dots, d_{nn})$  is the variance of the structural innovations.

But it also has a reduced form, which can be interpreted as a conventional VAR:

$$y_t = \Phi x_{t-1} + \varepsilon_t \quad (3)$$

$$\varepsilon_t \sim N(0_{n \times 1}, \Omega) \quad (4)$$

$$\Phi = A^{-1}B \quad (5)$$

$$\Omega = A^{-1}D(A^{-1})' \quad (6)$$

The model is estimated using Bayesian methods with sign restrictions. In set-identified models, the prior of the contemporaneous elasticities does not vanish asymptotically, as Baumeister and Hamilton (2015) (henceforth BH) demonstrated. The methodology was subsequently extended and applied to the global crude oil market by Baumeister and Hamilton (2019) and to the U.S. natural gas market by Rubaszek et al. (2021) and Farag (2024). The BH framework offers the flexibility to center all priors of the elements in  $\mathbf{A}$  at values supported by the literature. Anchoring the location and scale of the prior for the structural elasticities at empirically grounded values lets the model incorporate information that is not available from the data alone. In this section we provide a brief description of the methodology.

### 3.1 Prior

The starting point for setting the prior is decomposing  $p(A, B, D)$  as:

$$p(\mathbf{A}, \mathbf{B}, \mathbf{D}) = p(\mathbf{A}) \times p(\mathbf{D} \mid \mathbf{A}) \times p(\mathbf{B} \mid \mathbf{A}, \mathbf{D}) \quad (7)$$

The prior for the diagonal variance-covariance matrix  $\mathbf{D}$  consists in setting that the inverse of its elements  $d_{ii}^{-1}$  are jointly independent and follow a Gamma distribution with  $\kappa_i, \tau_i(\mathbf{A})$  as parameters.

$$p(\mathbf{D} \mid \mathbf{A}) = \prod_{i=1}^n p(d_{ii} \mid \mathbf{A}) \quad (8)$$

$$d_{ii}^{-1} \mid \mathbf{A} \sim \Gamma(\kappa_i, \tau_i(\mathbf{A})), \quad (9)$$

Similarly, rows of matrix  $\mathbf{B}$  are assumed to be jointly independent and to follow a multivariate normal distribution with parameters  $m_i, M_i$ .

$$p(\mathbf{B} \mid \mathbf{A}, \mathbf{D}) = \prod_{i=1}^n p(b_i \mid \mathbf{D}, \mathbf{A}) \quad (10)$$

$$b_i \mid \mathbf{A}, \mathbf{D} \sim N(m_i, d_{ii}M_i), \quad (11)$$

The prior for  $\mathbf{A}$  can be freely chosen to reflect any knowledge about the structural relations between the endogenous variables. In the case of this study it will be the joint distribution of several truncated Student-t.

### 3.2 Likelihood

Innovation of the structural model is assumed to be normal.

$$p(Y_T | \mathbf{A}, \mathbf{D}, \mathbf{B}) = (2\pi)^{-Tn/2} |\det(\mathbf{A})|^T \det(\mathbf{D})^{-T/2} \quad (12)$$

$$\times \exp \left\{ -\frac{1}{2} \sum_{t=1}^T (\mathbf{A}y_t - \mathbf{B}x_{t-1})' \mathbf{D}^{-1} (\mathbf{A}y_t - \mathbf{B}x_{t-1}) \right\}, \quad (13)$$

### 3.3 Posterior

To proceed with the posterior distribution, we need a similar decomposition as before:

$$p(\mathbf{A}, \mathbf{B}, \mathbf{D} | Y_t) = p(\mathbf{A} | Y_t) \times p(\mathbf{D} | \mathbf{A}, Y_t) \times p(\mathbf{B} | \mathbf{A}, \mathbf{D}, Y_t) \quad (14)$$

where the conditional posterior of the elements of the diagonal variance-covariance matrix,  $p(\mathbf{D} | \mathbf{A}, Y_t)$ , is expressed as the joint distribution of all elements on the diagonal:

$$p(\mathbf{D} | \mathbf{A}Y_t) = \prod_{i=1}^n p(d_{ii} | \mathbf{A}Y_t) \quad (15)$$

$$d_{ii}^{-1} | \mathbf{A} \sim \Gamma(\kappa_i^*, \tau_i^*(\mathbf{A})), \quad (16)$$

The parameters  $\kappa_i^*$  and  $\tau_i^*(\mathbf{A})$  that fully characterize this posterior are functions of the data and of  $\mathbf{A}$ , as follows:

$$\kappa_i^* = \kappa_i + T/2 \quad (17)$$

$$\tau_i^*(\mathbf{A}) = \tau_i(\mathbf{A}) + \xi_i^*(\mathbf{A}) \quad (18)$$

And the term  $\xi_i^*$  refers to the sum of squared residuals from regressing  $\tilde{Y}_i(\mathbf{A})$  on  $\tilde{X}_i$  defined as:

$$\xi_i^*(\mathbf{A}) = (\tilde{Y}_i(\mathbf{A})' \tilde{Y}_i(\mathbf{A})) - (\tilde{Y}_i(\mathbf{A})' \tilde{X}_i) (\tilde{X}_i' \tilde{X}_i)^{-1} (\tilde{X}_i' \tilde{Y}_i(\mathbf{A})) \quad (19)$$

$$\tilde{Y}_i = [y_1' a_i \quad \cdots \quad y_T' a_i \quad m_i' P_i]'_{(T+k) \times 1} \quad (20)$$

$$\tilde{X}_i = [x_0' \quad \cdots \quad x_{T-1}' \quad P_i]'_{(T+k) \times k} \quad (21)$$

where  $M_i = P_i' P_i$  and  $a_i$  are the rows of the matrix  $\mathbf{A}$ .

The conditional posterior of the autoregressive matrix,  $p(\mathbf{B} | \mathbf{A}, \mathbf{D}, Y_t)$ , can be defined as the joint posterior of its rows:

$$p(\mathbf{B} | \mathbf{A}, \mathbf{D}, Y_t) = \prod_{i=1}^n p(b_i | \mathbf{D}, \mathbf{A}) \quad (22)$$

$$b_i | \mathbf{A}, \mathbf{D}Y_t \sim N(m_i^*, d_{ii} M_i^*) \quad (23)$$

where

$$m_i^* = (\tilde{X}_i' \tilde{X}_i)^{-1} (\tilde{X}_i' \tilde{Y}_i), \quad (24)$$

$$M_i^* = (\tilde{X}_i' \tilde{X}_i)^{-1}. \quad (25)$$

Finally, the posterior of  $\mathbf{A}$  is:

$$p(\mathbf{A} \mid \mathbf{Y}_T) = \frac{k_T p(\mathbf{A}) [\det(\mathbf{A} \hat{\Omega}_T \mathbf{A}')]^{T/2}}{\prod_{i=1}^n [(2\tau_i^*/T)\kappa_i^*]} \prod_{i=1}^n \left\{ \frac{|\mathbf{M}_i^*|^{1/2}}{|\mathbf{M}_i|^{1/2}} \frac{\tau_i^{\kappa_i}}{\Gamma(\kappa_i)} \Gamma(\kappa_i^*) \right\}.$$

where  $\hat{\Omega}_T = T^{-1} \sum_{t=1}^T \hat{\varepsilon}_t \hat{\varepsilon}_t'$  is the sample variance covariance matrix estimated by maximum likelihood.

Inference draws from  $p(\mathbf{A} \mid \mathbf{Y}_T)$  using a Random-Walk Metropolis–Hastings sampler. The proposal density is a multivariate Gaussian with covariance  $\xi \cdot \hat{\mathcal{H}}^{-1}$ , where  $\hat{\mathcal{H}}$  is the Hessian of the log-posterior evaluated at the mode and the scaling factor is fixed at  $\xi = 0.0625$ . We run  $2 \times 10^6$  draws with a burn-in of  $1.7 \times 10^6$ , retaining the remaining  $3 \times 10^5$  post-burn-in draws for inference. Cumulative impulse responses, forecast-error variance decompositions, historical decompositions, and the counterfactual scenarios in Section 5.5 are computed for each retained draw and summarised by their pointwise posterior median and 5–95% quantiles. The random seed is fixed at 140778 for reproducibility. Convergence is assessed by visual inspection of the trace plots of the contemporaneous elasticities and by comparing the prior–posterior marginals reported in Section 5.1.

## 4 Empirical model and data

### 4.1 Data

We model the European natural gas market as a five-variable system that splits supply into two channels with markedly different short-run flexibility. The first channel, denoted  $q_{\text{pip}}$ , comprises net pipeline imports plus EU domestic production. Both share contractual and physical rigidities that prevent rapid adjustment within a month. Pipeline gas is governed by long-term, route-specific contracts with take-or-pay clauses, fixed transport infrastructure and limited spare capacity (Molnar, 2022). Domestic production is constrained by reservoir decline curves and regulatory frameworks that operate over multi-year horizons; the gradual phase-out of the Groningen field illustrates how the latter, rather than monthly price dynamics, drives variation in this margin. We aggregate the two because they share the relevant economic property for the question at hand—near-zero short-run price elasticity—which makes them the slow-adjusting margin of European gas supply.

The second channel,  $q_{\text{LNG}}$ , comprises net LNG imports. LNG operates as the flexible margin of European supply: cargoes are redirectable across regions in response to price differentials, contract durations have shortened, destination clauses have weakened, and exposure to spot prices has expanded (Sharples, 2023; Ritz, 2019). The 2022 energy crisis illustrates this asymmetry sharply: in response to the collapse in Russian pipeline deliveries, EU LNG imports surged by over 60% within months, while pipeline-and-domestic supply showed virtually no compensating expansion in the aggregate.

We deliberately keep  $q_{\text{pip}}$  as a single aggregate rather than introducing domestic production as a third VAR variable. The substantive distinction the model identifies is rigid versus flexible supply, and that distinction is economically binary: domestic production behaves

like pipeline imports (inelastic, slow-adjusting) and not like LNG. Adding a third supply variable would consume identification room without delivering economically distinct shocks, and the resulting series would be dominated by the Groningen-related structural decline rather than by short-run market dynamics. The robustness of this aggregation choice is verified in the appendix by re-estimating the model with  $q_{\text{pip}}$  restricted to pipeline imports only.

The dataset comprises five variables at monthly frequency. Pipeline-and-domestic supply  $q_{\text{pip}}$  and LNG imports  $q_{\text{LNG}}$  are constructed from the JODI-Gas database, summing reported pipeline imports, LNG imports and domestic production across 22 EU member states.<sup>1</sup> Total natural gas inventories  $s$  are also taken from JODI; the variable is included because of its central role in consumption smoothing and in precautionary or speculative behaviour. Industrial production  $ip$  is the manufacturing index for the euro area, sourced from Eurostat. The real price of natural gas  $p$  is the Dutch TTF benchmark from the World Bank Commodity Price Database, deflated by the euro area HICP.

$$y_t = [q_{\text{pip}}, ip, p, s, q_{\text{LNG}}] \quad (26)$$

All series are expressed in logarithmic growth rates (in percentage points per month) to remove unit roots in prices, industrial production and quantities. The sample spans February 2010 through the most recent month with complete JODI coverage; the pipeline, LNG, and inventory series are seasonally adjusted with X-13 ARIMA-SEATS, industrial production is sourced from Eurostat already seasonally adjusted, and the real gas price is left in raw form (not seasonally adjusted). The VAR is estimated with  $p = 6$  monthly lags, which is the longest specification permitted by the available degrees of freedom and is consistent with the lag choice in Baumeister and Hamilton (2019) and Rubaszek et al. (2021); the impulse responses are reported up to a horizon of 24 months.

## 4.2 Specification and assumptions

This section describes the structural model used in the paper to make the identification for the European natural gas market. The main characteristics of the contemporaneous elasticities among endogenous variables are equivalent to those used in Baumeister and Hamilton (2019); Rubaszek et al. (2021); Farag (2024).

We begin with the structural equations of the two supply components. Both pipeline and LNG imports are assumed to respond contemporaneously only to innovations in the gas price. Consistent with previous studies, both supply types are considered to be mildly elastic; that is, the corresponding supply price elasticities  $\gamma_{qp}$  and  $\beta_{qp}$  are assumed to differ

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<sup>1</sup>Country panel: AT, BE, CZ, DE, DK, EE, ES, FI, FR, GR, HU, IE, IT, LT, LU, LV, NL, PL, PT, SE, SI, SK. Five EU members are excluded due to incomplete JODI coverage over 2010–2024: BG, HR, MT and RO have data only from 2019 onwards, and CY does not report. The 22 countries jointly account for approximately 95% of EU-27 natural gas consumption. For the inventory series (see below), NL is additionally dropped because both its closing-stock-level and stock-change reports start only in January 2014; Germany’s pre-2013 inventory level is reconstructed by backward-extrapolating from January 2013 using monthly aggregate stock changes, which DE reports throughout the sample. A robustness check using IEA *Gas Trade Flows* data is left for a subsequent revision.

from zero. Because the production and delivery of natural gas involve adjustment costs, imports are expected to react to real activity shocks only with a delay of at least one month. Moreover, since the output of these two technologies is extracted simultaneously, they cannot be adjusted relative to one another within the same period.

$$q_t^{pip} = \gamma_{qp}p_t + \mathbf{b}'_1x_{t-1} + \varepsilon_t^{\text{pipeline supply}} \quad (27)$$

$$q_t^{LNG} = \beta_{qp}p_t + \mathbf{b}'_5x_{t-1} + \varepsilon_t^{\text{LNG supply}} \quad (28)$$

Real activity is broadly proxied by industrial production, which is assumed to respond contemporaneously only to the gas price. This assumption is motivated by the fact that a supply disruption affects industrial activity primarily through its impact on prices. In the short run, however, a shortfall in gas supply can be partially offset by substituting other components of the energy mix, such as biogas or alternative electricity sources. Nevertheless, any increase in the gas price is expected to exert a negative effect on industrial production.

$$ip_t = \alpha_{yp}p_t + \mathbf{b}'_2x_{t-1} + \varepsilon_t^{\text{aggregate demand}} \quad (29)$$

Demand for natural gas is separated into *flow demand* and *precautionary demand*. Flow demand captures shifts in the portion of gas that is effectively consumed, and can be associated with changes in market conditions that make gas more desirable to consumers. The structural demand equation incorporates two contemporaneous channels: industrial production affects gas demand through the income elasticity  $\alpha_{qy} > 0$ , and the gas price affects demand through the price elasticity  $\alpha_{qp} < 0$ . Since consumers are indifferent to the source of natural gas supply, any unit of pipeline gas, LNG, or stored gas is treated as identical, so aggregate demand is defined as  $Q_t = q_t^{pip} + q_t^{LNG} - s_t$ . In this case, the structural demand equation is given by:

$$Q_t = \alpha_{qy}ip_t + \alpha_{qp}p_t + \mathbf{b}'_3x_{t-1} + \varepsilon_t^{\text{flow demand}}, \quad (30)$$

or equivalently, in inverse demand form:

$$\alpha_{qp}p_t = q_t^{pip} + q_t^{LNG} - s_t - \alpha_{qy}ip_t - \mathbf{b}'_3x_{t-1} - \varepsilon_t^{\text{flow demand}}. \quad (31)$$

Precautionary demand captures changes in market participants' expectations that lead to adjustments in natural gas inventories. Such behavior may arise either because agents anticipate a potential supply disruption or because a rise in prices triggers speculative motives among inventory holders. Stocks of gas are adjusted in response to current imports, making inventories a potential substitute for them. Inventories also react to price innovations within the same month, because any shock that raises the gas price has two competing effects on stocks. On one hand, a higher price gives inventory holders the opportunity to realize a profit by selling at a price above their acquisition cost, implying that stocks decline as the price rises ( $\psi_2 < 0$ ). On the other hand, if inventory holders interpret the current price increase as a precursor of further price hikes, they may prefer to expand current inventories in order to sell more gas in the future ( $\psi_2 > 0$ ). These two conflicting forces make the sign of the price elasticity  $\psi_2$  theoretically ambiguous, and thus, like the other two inventory parameters  $\psi_1$  and  $\psi_3$ , it is left without a sign restriction.

$$s_t = \psi_1q_t^{pip} + \psi_2p_t + \psi_3q_t^{LNG} + \mathbf{b}'_4x_{t-1} + \varepsilon_t^{\text{precautionary demand}} \quad (32)$$

Table 1: Contemporaneous elasticities in  $\mathbf{A}$ 

| Parameter     | Meaning                                | Sign restriction  |
|---------------|----------------------------------------|-------------------|
| $\gamma_{qp}$ | Pipeline supply elasticity             | $\gamma_{qp} > 0$ |
| $\beta_{qp}$  | LNG supply elasticity                  | $\beta_{qp} > 0$  |
| $\alpha_{yp}$ | Effect of $p$ on economic activity     | $\alpha_{yp} < 0$ |
| $\alpha_{qy}$ | Income elasticity of gas demand        | $\alpha_{qy} > 0$ |
| $\alpha_{qp}$ | Gas demand elasticity                  | $\alpha_{qp} < 0$ |
| $\psi_1$      | Effect of $q^{pip}$ on gas inventories | none              |
| $\psi_2$      | Effect of $p$ on gas inventories       | none              |
| $\psi_3$      | Effect of $q^{LNG}$ on gas inventories | none              |

### 4.3 Matrix form

Following the specification of a SVAR the model is rewritten in matrix form as it follows:

$$\begin{pmatrix} 1 & 0 & -\gamma_{qp} & 0 & 0 \\ 0 & 1 & -\alpha_{yp} & 0 & 0 \\ 1 & -\alpha_{qy} & -\alpha_{qp} & -1 & 1 \\ -\psi_1 & 0 & -\psi_2 & 1 & -\psi_3 \\ 0 & 0 & -\beta_{qp} & 0 & 1 \end{pmatrix} \begin{pmatrix} q_t^{pip} \\ ip_t \\ p_t \\ s_t \\ q_t^{LNG} \end{pmatrix} = \mathbf{B}x_{t-1} + \begin{pmatrix} u_t^{\text{pipeline supply}} \\ u_t^{\text{aggregate demand}} \\ u_t^{\text{flow demand}} \\ u_t^{\text{precautionary shock}} \\ u_t^{\text{LNG supply}} \end{pmatrix}$$

Where the structural shocks are assumed to be normally distributed  $u_t \sim N(0_{n \times 1}, \mathbf{D})$  and the contemporaneous elasticities in  $\mathbf{A}$  satisfy the following five sign restrictions:

$$\gamma_{qp} > 0 \quad \beta_{qp} > 0 \quad \alpha_{yp} < 0 \quad \alpha_{qy} > 0 \quad \alpha_{qp} < 0$$

The way to implement these restrictions is to incorporate them in the prior of the contemporaneous matrix,  $p(\mathbf{A})$ , assigning truncated distributions in each case. The purpose of this is twofold, first it makes easier to incorporate prior knowledge of the elements in  $\mathbf{A}$  that is not directly provided by the likelihood of the data. On top of that it helps to ensure that the posterior of the elasticities  $p(\mathbf{A} \mid Y_t)$  gives most of the probability to values of  $\mathbf{A}$  that make the impact matrix  $\mathbf{A}^{-1}$  to satisfy the following signs.

$$\begin{pmatrix} \varepsilon_{q^{pip},t} \\ \varepsilon_{ip,t} \\ \varepsilon_{p,t} \\ \varepsilon_{s,t} \\ \varepsilon_{q^{LNG},t} \end{pmatrix} = \begin{pmatrix} + & + & + & ? & ? \\ + & + & - & ? & + \\ - & + & + & + & - \\ ? & ? & + & ? & ? \\ ? & + & + & ? & + \end{pmatrix} \begin{pmatrix} u_t^{\text{pipeline supply}} \\ u_t^{\text{aggregate demand}} \\ u_t^{\text{flow demand}} \\ u_t^{\text{precautionary shock}} \\ u_t^{\text{LNG supply}} \end{pmatrix}$$

### 4.4 Prior knowledge about these elasticities

The structural elasticities embedded in matrix  $A$ , which govern the contemporaneous interactions between quantities, prices, and real economic activity, can be disciplined using a well-established empirical literature on energy markets, particularly natural gas and oil. In set-identified SVAR models that rely on inequality or sign restrictions, these elasticities play a central role in shaping inference. Unlike in fully identified models, the posterior

distribution does not converge to the likelihood even asymptotically, implying that prior information does not wash out with increasing sample size (Baumeister and Hamilton, 2015). As a consequence, the choice of prior distributions is not innocuous and must be grounded in economically and empirically plausible values.

The short-run price elasticity of pipeline natural gas supply, denoted by  $\gamma_{qp}$ , measures the responsiveness of pipeline gas imports to price changes. We impose the theoretical restriction  $\gamma_{qp} > 0$  and center the prior at 0.1, reflecting the well-established view that natural gas supply is highly inelastic in the short run. Empirical studies consistently find small elasticities, typically close to zero or below 0.1, due to technological constraints, adjustment costs, and contractual rigidities (Al-Sahlawi, 1989; Dahl and Duggan, 1996; Arora, 2014; Rubaszek et al., 2021; Mason and Roberts, 2018). Centering the prior at a small positive value therefore captures an upward-sloping but steep supply curve, while the heavy-tailed Student- $t$  prior allows for uncertainty without admitting implausibly large elasticities.

A similar prior is imposed on the short-run price elasticity of LNG supply,  $\beta_{qp}$ . Although LNG-specific elasticity estimates are scarce, short-run LNG supply is constrained by fixed liquefaction capacity, shipping availability, and regasification infrastructure, implying only limited responsiveness to price signals. We therefore impose  $\beta_{qp} > 0$  and center the prior at 0.1, allowing the data to assess whether LNG supply is more flexible than pipeline supply while anchoring inference in a plausible range.

On the demand side, the short-run price elasticity of natural gas consumption,  $\alpha_{qp}$ , is expected to be negative. We center the prior at  $-0.3$  and truncate the distribution to enforce  $\alpha_{qp} < 0$ . This calibration reflects the broad consensus that gas demand is highly inelastic in the short run. Early evidence summarized by Al-Sahlawi (1989) documents short-run elasticities clustered around values close to  $-0.3$ , while more recent studies typically estimate elasticities between  $-0.04$  and  $-0.16$  (Joutz, 2009; Bernstein and Madlener, 2011; Arora, 2014), with demand remaining far from elastic even when allowing for heterogeneity (Burke and Yang, 2016).

The elasticity of gas demand with respect to aggregate economic activity,  $\alpha_{qy}$ , captures the income elasticity of natural gas demand. We impose  $\alpha_{qy} > 0$  and center the prior at 0.7, reflecting a strong but less than proportional response of gas consumption to changes in output. This calibration is motivated by survey evidence summarized by Al-Sahlawi (1989), which reports income elasticities ranging from near zero to above unity, as well as by more recent cross-country estimates for OECD economies that cluster around 0.7 Burke and Yang (2016). Anchoring the prior at this benchmark provides an empirically grounded characterization of demand responsiveness to economic activity.

Finally, the parameter  $\alpha_{yp}$  captures the contemporaneous response of aggregate economic activity to changes in natural gas prices. We impose a negative sign restriction and center the prior at  $-0.05$ , reflecting the well-established view that higher energy prices act as a drag on output by increasing production costs and reducing real disposable income, while having only a modest immediate effect Kilian (2008). Similar calibrations are standard in Bayesian SVAR studies of energy markets, and posterior estimates are typically close to zero Baumeister and Hamilton (2019). Evidence from the European gas crisis likewise points to

Table 2: Priors and posteriors distributions for parameters in **A**

|                                                               | $\gamma_{qp}$ | $\beta_{qp}$ | $\alpha_{yp}$ | $\alpha_{qy}$ | $\alpha_{qp}$ | $\psi_1$ | $\psi_2$ | $\psi_3$ |
|---------------------------------------------------------------|---------------|--------------|---------------|---------------|---------------|----------|----------|----------|
| <i>Priors affecting contemporaneous coefficients <b>A</b></i> |               |              |               |               |               |          |          |          |
| Location                                                      | 0.1           | 0.1          | -0.05         | 0.7           | -0.3          | 0        | 0        | 0        |
| Scale                                                         | 0.2           | 0.2          | 0.1           | 0.3           | 0.1           | 0.5      | 0.5      | 0.5      |
| Degrees of freedom                                            | 3             | 3            | 3             | 3             | 3             | 3        | 3        | 3        |
| Sign restriction                                              | $t^+$         | $t^+$        | $t^-$         | $t^+$         | $t^-$         | $t$      | $t$      | $t$      |
| <i>Posterior of contemporaneous coefficients <b>A</b></i>     |               |              |               |               |               |          |          |          |
| 5%                                                            | 0.034         | 0.825        | -0.028        | 0.446         | -1.098        | -0.157   | -0.108   | -0.011   |
| 50%                                                           | 0.079         | 1.629        | -0.009        | 0.894         | -0.530        | -0.041   | -0.067   | 0.030    |
| 95%                                                           | 0.125         | 3.028        | -0.001        | 1.535         | -0.283        | 0.073    | -0.026   | 0.072    |
| Mean                                                          | 0.079         | 1.745        | -0.011        | 0.929         | -0.592        | -0.042   | -0.067   | 0.030    |

noticeable but non-catastrophic short-run output effects Güntner et al. (2024), supporting a small negative prior mean.

In summary, all prior distributions are centered on values firmly supported by the empirical literature. Supply elasticities are assumed to be small and positive, demand price elasticities moderate and negative, income elasticities positive and sizable, and the contemporaneous output response to gas prices modestly negative. By adopting literature-based priors within the Bayesian framework of Baumeister and Hamilton (2015), the model achieves economically meaningful identification while avoiding improper posteriors. Although the data remain free to update these beliefs, the priors provide a necessary anchor that ensures sensible inference in a weakly identified SVAR setting.

## 5 Results

In this section, we present the estimation results of the baseline model and discuss the dynamics of the European natural gas market. We begin by examining the posterior distributions of the model parameters, focusing in particular on the short-run price elasticities of demand and supply. Then, we analyze the posterior impulse response functions, which capture the dynamic responses of the endogenous variables to structural shocks. Next, we quantify both the short- and long-term effects of these shocks on the variables included in the VAR system and assess their historical decompositions, highlighting their contributions to the variation in natural gas prices. Finally, we conduct a counterfactual analysis based on a hypothetical scenario in which European LNG imports were not available during the post-energy-crisis recovery period.

## 5.1 Posterior knowledge of the elasticities

We begin by comparing the prior and posterior distributions of the parameters in the contemporaneous relations matrix  $\mathbf{A}$ , with particular attention to the posterior medians of the short-run demand and supply elasticities. Figure 1 displays the prior distributions (red line) alongside the posterior ones (blue histograms), while the corresponding descriptive statistics are reported in the lower panel of Table 2. The main findings can be summarized as follows.

The posterior median of the short-run price elasticity of pipeline-and-domestic supply,  $\gamma_{qp}$ , is estimated at only 0.079, indicating a highly inelastic response of the rigid supply margin to price changes. This estimate is broadly consistent with the elasticities reported by Rubaszek et al. (2021) and Farag (2024) for the U.S. natural gas market, who find elasticities of approximately 0.01 and 0.02, respectively, and conclude that American gas supply is strongly price inelastic. It is also of the same order of magnitude as the elasticity estimated by Mason and Roberts (2018) for natural gas wells in Wyoming, who report a value of 0.026. Moreover, our estimate is comparable to the European supply elasticities obtained by Adolfsen et al. (2024) and to those reported by Güntner et al. (2024) for the German natural gas market, further confirming the limited short-run responsiveness of contracted-and-domestic gas supply to price movements.

The price elasticity of LNG supply,  $\beta_{qp}$ , tells a sharply different story. Its posterior median is 1.63, with a 90% credible interval of  $[0.83, 3.03]$  that excludes any value below unity. LNG supply is therefore estimated to be highly responsive to price: a one-percent increase in the real gas price is associated with a contemporaneous expansion of LNG imports of roughly 1.6%. The contrast with  $\gamma_{qp} \approx 0.08$  is striking—one full order of magnitude—and provides the clearest empirical support for the rigid-versus-flexible characterisation that motivates the distinction between the two supply channels. The estimate is consistent with the institutional view of LNG as the marginal balancing source in Europe: cargoes can be redirected across regions in response to price differentials, contract durations have shortened, and exposure to spot pricing has expanded (Sharples, 2023; Ritz, 2019).

Regarding demand elasticities, the posterior median of the short-run price elasticity,  $\alpha_{qp}$ , is estimated at  $-0.53$ , with a 90% credible interval of  $[-1.10, -0.28]$ . In absolute terms this value is moderately larger than the estimates reported by Rubaszek et al. (2021) and Farag (2024) for the U.S. natural gas market ( $-0.42$  and  $-0.18$ , respectively). The result suggests that European demand is somewhat more responsive to price than its U.S. counterpart, a finding consistent with the larger role of fuel-switching in electricity generation and with the greater exposure of European industrial users to spot pricing rather than long-term take-or-pay contracts. The estimate is also broadly in line with the upper end of the cross-country range surveyed by Burke and Yang (2016).

As for the income elasticity,  $\alpha_{qy}$ , its posterior distribution remains close to the prior, with a median of 0.89, broadly consistent with the 0.719 reported by Güntner et al. (2024) for the German market and the 0.725 obtained by Baumeister and Hamilton (2019) for crude oil. This similarity indicates that shifts in aggregate economic activity translate into comparable changes in demand across both energy commodities and across different markets.

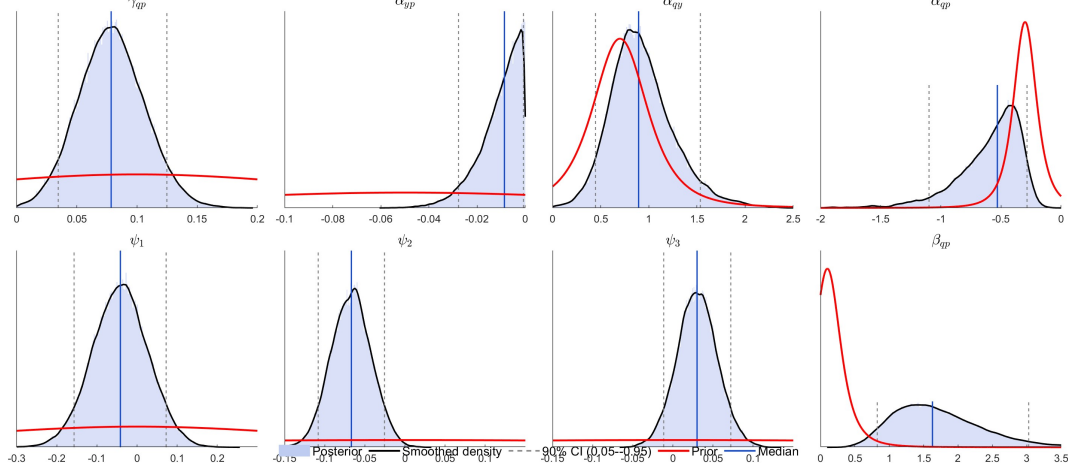


Figure 1: Prior and posterior of contemporaneous elasticities for the baseline model. Each panel reports one element of the contemporaneous matrix  $\mathbf{A}$ :  $\gamma_{qp}$  (pipeline supply elasticity),  $\alpha_{yp}$  (industrial-activity response to price),  $\alpha_{qy}$  (demand income elasticity),  $\alpha_{qp}$  (demand price elasticity),  $\psi_1$ ,  $\psi_2$  and  $\psi_3$  (storage equation, capturing the effects of pipeline supply, price, and LNG supply on inventories), and  $\beta_{qp}$  (LNG supply elasticity). The light-blue filled histogram shows the marginal posterior; the solid black line is the corresponding smoothed (kernel) density; the vertical dashed grey lines mark the 5% and 95% posterior quantiles (90% credible interval); the solid vertical blue line marks the posterior median; the solid red line is the prior density used for the parameter (see the prior specification in Section 4.4 and the legend at the bottom).

For the remaining parameters, the estimates indicate that, in the short run, the effect of natural gas prices on economic activity,  $\alpha_{yp}$ , is almost negligible, with a posterior median of  $-0.009$ . This value is well below both our prior belief and the estimates reported by Adolfsen et al. (2024) for the euro area, yet it is close to the  $-0.002$  obtained by Baumeister and Hamilton (2019) for the crude oil market.

Turning to the inventory equation, the contemporaneous effects of pipeline-and-domestic supply ( $\psi_1$ ), prices ( $\psi_2$ ), and LNG imports ( $\psi_3$ ) on inventories show modest magnitudes but distinct patterns. The posterior median of  $\psi_2$  is  $-0.067$ , with a 90% credible interval entirely below zero ( $[-0.108, -0.026]$ ): a higher contemporaneous price reduces inventories, consistent with stockholders releasing gas to capture the price gain rather than accumulating inventories in anticipation of further increases. The posterior median of  $\psi_1$  is small and not significantly different from zero ( $-0.041$ ,  $[-0.157, 0.073]$ ). By contrast,  $\psi_3$  has a posterior median of  $+0.030$ , with a 90% credible interval that almost entirely lies above zero ( $[-0.011, 0.072]$ ): positive innovations to LNG imports translate contemporaneously into higher inventories. The opposite signs of  $\psi_1$  and  $\psi_3$  are economically intuitive in light of the institutional differences between supply channels: rigid (pipeline-and-domestic) gas is largely delivered directly into the system, whereas LNG cargoes are typically buffered in regasification and storage facilities before distribution.

## 5.2 Impulse Response Functions

We decompose the dynamics of the European natural gas market into five structural shocks, following the structural oil market SVAR literature Kilian (2009); Kilian and Murphy (2012, 2014); Baumeister and Hamilton (2019). The model distinguishes two supply shocks, pipeline gas and LNG, and three demand shocks, aggregate demand, flow demand, and precautionary or storage demand. We trace their cumulative impulse responses for the set of endogenous variables.

First, consider the IRFs to a pipeline supply shock. An adverse pipeline shock lowers pipeline imports on impact and raises the real natural gas price, while industrial production declines in the short run. Inventories fall for a few months, consistent with an initial drawdown that partially cushions the physical shortfall, and LNG imports increase as supply is reallocated toward alternative inflows. In our estimates, the price effect is transitory and dissipates within about five months after the shock.

Next, consider the IRFs to a positive LNG supply shock that raises total gas availability, leading to a sustained decline in the real natural gas price and a persistent improvement in market conditions. Inventories increase for an extended period as the additional supply supports storage accumulation, while the adjustment in other inflows is gradual. Conversely, an adverse LNG shock reduces LNG availability, generates a prolonged tightening of the market, and raises prices in a persistent manner, reflecting the central role of LNG as the medium-run balancing margin in Europe.

These dynamics align with the gas literature documenting temporary price effects of pipeline disruptions. For example, Adolfsen et al. (2024) also find that pipeline supply shocks raise natural gas prices only temporarily, although their price response persists for up to nine months. A natural explanation is that their supply shock aggregates pipeline and LNG supply, so the estimated response reflects a mixture of short-lived pipeline disruptions and more persistent LNG-related adjustment. By distinguishing the two supply channels, our framework isolates the shorter-lived nature of pipeline shocks more clearly. The European experience in 2022, marked by the sharp reduction in Russian pipeline deliveries, provides a salient illustration of this mechanism, with pronounced price increases accompanied by comparatively limited immediate output losses as inventories and substitute imports partially mitigated the supply shortfall.

Compared with pipeline shocks, LNG shocks tend to be more persistent because LNG is the marginal balancing source in Europe and therefore shapes the medium-run market-clearing price. A pipeline disruption is typically a corridor-specific shortfall whose price impact is partly offset through inventory withdrawals, demand adjustment, and substitution toward other inflows, so the price effect can fade relatively quickly. An LNG supply shock instead reflects a change in Europe’s access to flexible global cargoes, which relaxes the constraint on replacement supply and shifts the marginal supply schedule for longer. As a result, LNG shocks affect prices more durably than pipeline shocks in the impulse responses.

Turning to aggregate and flow demand shocks (second and third columns), both generate upward pressure on natural gas prices, but they differ in the persistence of the price re-

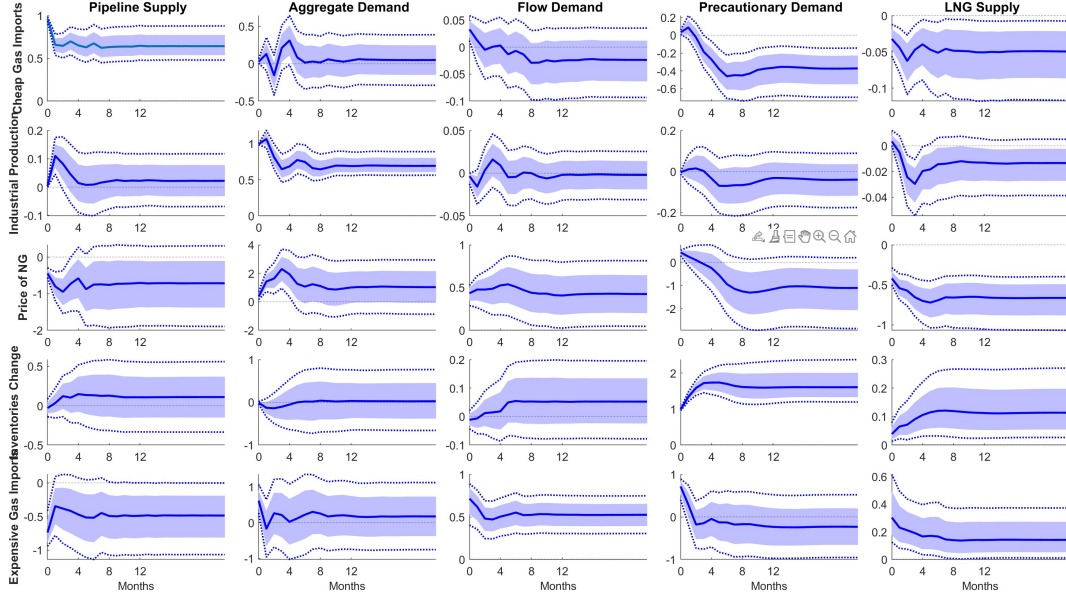


Figure 2: Cumulative Impulse Responses of the endogenous variables (rows) to the five structural shocks (columns). The solid blue line is the posterior median, the light-blue shaded area is the 68% credible band, and the dotted blue lines are the 90% credible band. The horizon is 12 months. All responses are cumulated. Row labels denote the responding variable; column titles denote the identified shock.

sponse. Aggregate demand shocks raise prices only temporarily, consistent with a gradual adjustment of quantities as the economy reverts toward baseline. Flow demand shocks, in contrast, trigger a sharper contemporaneous tightening of market conditions, reflected in an immediate price increase and a pronounced drawdown of inventories, and they may be mildly contractionary for industrial production as higher energy costs and short-run scarcity constrain activity. Over time, however, the price effect of flow demand shocks fades as consumption normalizes and inventories are replenished.

Finally, precautionary demand shocks capture shifts in desired inventory holdings driven by expectations of future supply scarcity. A positive precautionary shock induces a simultaneous increase in inventories and natural gas prices, consistent with stockpiling behavior. In response, LNG imports remain broadly unchanged, whereas pipeline imports decline and settle at a permanently lower level after about three months. This pattern is suggestive of forward-looking inventory management, whereby storage holders anticipate persistent disruptions in pipeline supply, as in the 2022 episodes, and adjust their sourcing and inventory decisions accordingly.

Overall, the cumulative impulse responses indicate that European natural gas prices reflect the interaction of demand conditions, supply rigidities, and storage behavior. In the short run, price fluctuations are driven by both supply- and demand-side shocks, whereas medium-run dynamics are mostly associated with flow demand and LNG supply adjustments. These patterns are consistent with evidence from the oil and gas literatures and highlight the particular sensitivity of gas prices to infrastructure constraints and concerns about supply

Table 3: Median FEVD of the Price

|          | <b>Pipeline Sup.</b> | <b>Aggr. Dem.</b> | <b>Flow Dem.</b> | <b>Prec. Dem.</b> | <b>LNG Sup.</b> |
|----------|----------------------|-------------------|------------------|-------------------|-----------------|
| 1 month  | 22.1                 | 18.1              | 20.3             | 20.3              | 19.1            |
| 2 months | 16.9                 | 53.8              | 8.6              | 12.1              | 8.6             |
| 3 months | 17.3                 | 54.1              | 7.6              | 13.5              | 7.5             |
| 6 months | 15.6                 | 61.4              | 4.5              | 14.0              | 4.5             |
| 1 year   | 14.8                 | 59.6              | 3.9              | 17.9              | 3.9             |
| 2 years  | 14.7                 | 59.5              | 3.9              | 18.1              | 3.8             |

Notes: Median forecast error variance decomposition (in percent) of the natural gas real price at horizons of 1, 2, 3, 6, 12 and 24 months. Rows do not necessarily sum to 100 because the table reports posterior medians, not posterior shares of a single draw. Data are monthly, 2010–2024.

security. The horizon-dependent structure also carries a direct implication for monetary policy. Because pipeline-supply shocks transmit to gas prices as relatively short-lived spikes, they are likely to translate into transient headline-inflation deviations that a central bank can credibly look through; LNG-supply shocks, by contrast, propagate more persistently and may therefore exert pressure on medium-run inflation expectations and on core measures through second-round effects, consistent with the gas-to-inflation pass-through documented for the euro area by Adolfsen et al. (2024).

### 5.3 Variance Decomposition

A Forecast Error Variance Decomposition (FEVD) quantifies the relative importance of each structural shock in explaining the uncertainty of gas price fluctuations over time, thereby providing a summary measure of the dominant drivers of price variability. Table 3 reports the median forecast error variance decomposition (FEVD) of the natural gas price at different horizons.

At very short horizons, price fluctuations are driven by a relatively balanced mix of all the shocks in the model. At the one-month horizon, the contributions range from 18.1 percent for aggregate demand to 22.1 percent for pipeline supply, with flow demand, precautionary demand and LNG supply each accounting for roughly 20 percent. This pattern reflects the fact that in the immediate aftermath of shocks, gas prices react to a wide range of disturbances affecting both supply and demand conditions. As the horizon increases, aggregate demand shocks emerge as the dominant source of price variance. From two months onward, aggregate demand explains more than 50 percent of the forecast error variance, rising above 60 percent at the six-month horizon. This indicates that medium- to long-run movements in natural gas prices are strongly linked to broader macroeconomic conditions and demand forces, rather than to short-lived supply shocks.

Turning to the relative roles of supply and demand shocks, the variance decomposition reveals clear differences in their persistence and horizon dependence. On the demand side, flow demand shocks, which capture short-run adjustments in industrial and residential consumption, play an important role at very short horizons, accounting for more than 20 percent of

price variance at the one-month horizon. Their contribution, however, declines rapidly and falls below 5 percent beyond one year, indicating that immediate consumption adjustments have largely transitory effects on price volatility. In contrast, precautionary demand shocks display a more persistent pattern. While their short-run contribution is moderate, their importance increases with the forecast horizon, explaining close to 20 percent of price variance at one- and two-year horizons. Unlike flow demand shocks, which dissipate once short-run consumption adjusts, precautionary motives affect prices through expectations and inventory decisions that operate over longer horizons. This suggests that uncertainty-driven behavior and expectations about future supply conditions generate lasting effects on prices, in line with the evidence from the energy crisis.

On the supply side, both pipeline and LNG supply shocks contribute more modestly to overall price variance when averaged over the full sample. Pipeline supply shocks explain a non-negligible share of short-run price variability but stabilize at around 15 percent at longer horizons, pointing to a limited yet persistent influence. Meanwhile, LNG supply shocks account for an even smaller fraction of price variance, falling to below 5 percent at horizons beyond six months. This indicates that while supply disruptions, particularly in LNG markets, can trigger sharp price movements during specific episodes, their contribution to overall price uncertainty is relatively limited in the long run.

Taken together, these results highlight a clear horizon-dependent structure in which short-run price fluctuations reflect a combination of supply disturbances and immediate demand adjustments, whereas medium- and long-run volatility is increasingly shaped by aggregate demand conditions and persistent precautionary shocks rather than by supply-side forces.

## 5.4 Historical Decomposition

A historical decomposition expresses the evolution of an observed time series as the cumulative contribution of the structural shocks identified in the model. In a structural VAR framework, this tool is particularly useful for attributing large price movements to distinct economic mechanisms and for assessing how the relative importance of supply, demand, and precautionary forces varies over time. Rather than focusing on the response to isolated shocks, the historical decomposition provides a narrative interpretation of realized outcomes, allowing one to identify which shocks were most relevant during specific historical episodes.

The dynamics of European natural gas real prices are best understood as the outcome of three interacting forces: (i) very low short-run demand elasticity, (ii) binding supply and infrastructure constraints, and (iii) the pivotal role of storage and expectations when supply security deteriorates. Because gas is hard to substitute in heating and many industrial processes in the short run, even moderate physical imbalances translate into large price movements. In Europe, these imbalances are magnified by the fact that pipeline supply is tied to specific corridors and contracts, while LNG is globally traded but constrained by liquefaction capacity, shipping availability, and regasification infrastructure. Hence, price spikes emerge when the market must simultaneously (a) replace lost pipeline volumes, and (b) rebuild inventories ahead of winter, especially under heightened uncertainty. Figure 3 presents the historical decomposition of the European real natural gas price based on the

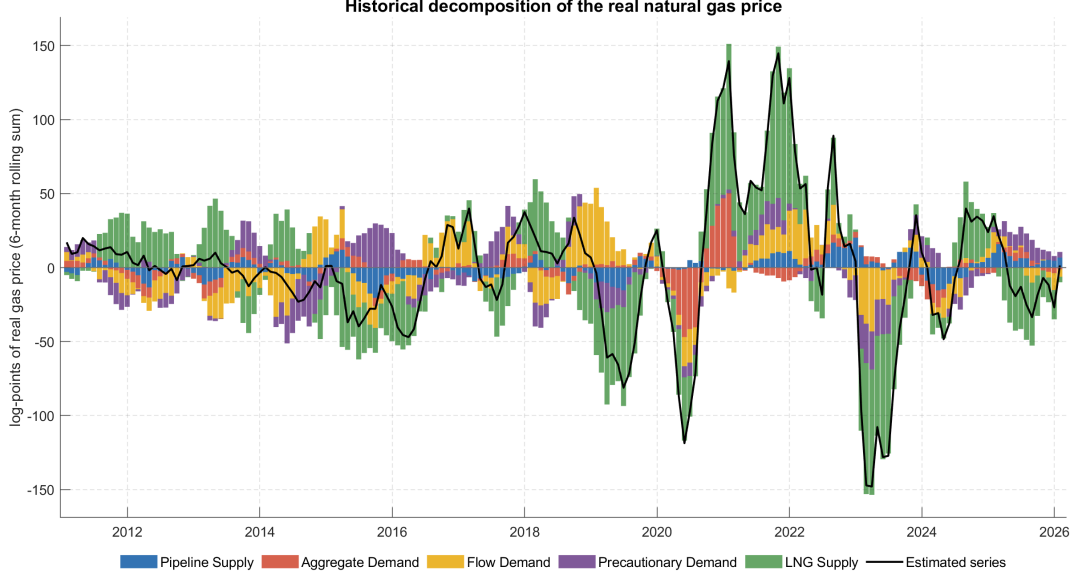


Figure 3: Historical decomposition of the European real natural gas price. Stacked coloured bars show the cumulative six-month contribution of each of the five identified structural shocks (steel blue: pipeline supply; terracotta: aggregate demand; mustard: flow demand; muted purple: precautionary/storage demand; sage green: LNG supply). The solid black line is the model-implied estimated series (sum of all shock contributions). Values are in log-points of the real gas price, plotted at monthly frequency over the period 2010–2024.

five structural shocks identified in the model. The black line depicts the estimated log-level price, while the colored bars show the contemporaneous contribution of pipeline supply, aggregate demand, flow demand, precautionary demand, and LNG supply shocks. Several episodes align closely with well-documented market developments. Following the 2014 Crimea invasion and the deterioration in EU Russia energy relations, the decomposition suggests an increase in storage-related and flow pressures, although price effects remained contained, consistent with the gradual nature of the adjustment and the buffering role of contracts and infrastructure. The COVID-19 shock in 2020 appears as a demand-driven episode: the sharp decline in prices coincides with large negative contributions from flow demand and precautionary demand, consistent with the collapse in economic activity and energy use documented for global gas markets during the pandemic International Energy Agency (2020). In the post-COVID recovery of 2021, prices start rising well before the full-scale 2022 crisis. This pattern is consistent with the combination of a rapid rebound in demand and supply-side frictions in global LNG markets and shipping <sup>2</sup>, together with unusually low European storage entering the winter season Popkostova (2022). The most pronounced regime shift occurs during the 2022 energy crisis, when the sudden reduction in Russian pipeline deliveries forced Europe to rely on scarce flexible LNG volumes under tight global capacity conditions International Energy Agency (2023). In this period, the decom-

<sup>2</sup>Specially a stronger power and gas demand in Asia pulled LNG away from Europe Reuters (2021). This has combined with a global supply crunch and limited injections, resulting in record-low stocks in the second half of 2021

Table 4: Summary of the three counterfactual scenarios

| Scenario                           | Constraint                             | Window          | Median gap | 90% CI         | Ratio          |
|------------------------------------|----------------------------------------|-----------------|------------|----------------|----------------|
| CF1 LNG frozen at Feb-22           | $g_t^{LNG} = 0$                        | Feb-22 – Dec-23 | +0.80      | [0, +2.3]      | $\times 2.20$  |
| CF2 LNG absorbs pipeline shortfall | $\Delta q_t^{LNG} = -\Delta q_t^{pip}$ | Feb-22 – Dec-23 | -2.00      | [-7, -1]       | $\times 0.135$ |
| CF3 No autonomous demand reduction | $u_t^3 = 0$                            | Aug-22 – Mar-23 | +0.25      | [-0.05, +0.75] | $\times 1.28$  |

Notes: Median price gap is the posterior median of  $\log(p_t^{CF}) - \log(p_t)$  evaluated at the end of the sample, expressed in log-points; the 90% credible interval is the corresponding [5, 95] posterior range. Ratio is  $\exp(\text{median gap})$  and reports the implied counterfactual level relative to the observed series (e.g. a ratio of  $\times 2.20$  means the counterfactual price is 2.2 times the observed one). Counterfactuals 1 and 2 are framed as polar bounds on what the LNG margin could have delivered (upper and lower bounds respectively); Counterfactual 3 measures the joint contribution of the autonomous demand reduction over the formal EU policy window.

position is dominated by positive contributions from flow demand, precautionary demand, and LNG supply shocks, consistent with scarcity concerns and storage motives playing a central role in price formation. Aggregate demand contributes comparatively little during the spike, which is also consistent with contemporaneous assessments that emphasize supply insecurity and risk premia rather than macroeconomic overheating Furtuna et al. (2022). Finally, the normalization in 2023 coincides with the unwinding of these forces and with improved fundamentals, including strong storage buffers and the gradual expansion of LNG import capacity and market resilience EuropeanCommission (2024).

Overall, the historical decomposition clarifies how the same structural mechanisms operate with very different intensity across episodes. Periods of stability are characterized by offsetting shocks that keep prices anchored, whereas crisis episodes arise when multiple supply- and risk-related forces reinforce each other. The analysis shows that large price movements emerge not from a single dominant shock, but from the interaction between constrained supply, expectations, and limited adjustment margins. This perspective helps rationalize both the severity of the 2022 price spike and the speed of its subsequent reversal.

## 5.5 Counterfactual Analysis

The structural VAR framework employed in this paper enables a rich set of counterfactual and scenario-based analyses, in line with Güntner et al. (2024) and Antolín-Díaz et al. (2021). We construct three different counterfactual scenarios to evaluate the role of LNG and demand adjustment in shaping the price trajectory during the 2022–2023 European energy crisis. Table 4 summarises the headline results before each scenario is discussed in turn.

We follow the convention of Antolín-Díaz et al. (2021) and construct counterfactuals at the level of the observed endogenous variables (a *data* counterfactual) rather than by zeroing out subsets of structural shocks. Concretely, for each counterfactual scenario we impose an alternative path on one of the endogenous variables over the crisis window, re-run the historical decomposition on the modified dataset, and reconstruct the implied path of the real gas price by accumulating the resulting monthly log-growth rates from January 2022. This approach is well suited to the questions we ask: each scenario is naturally framed in

terms of a counterfactual time path for an observed margin of adjustment (LNG imports, flow demand) rather than in terms of an isolated structural innovation. The first scenario asks how prices would have evolved had LNG imports not been able to expand above their pre-crisis level; the second asks how prices would have responded under a more aggressive demand-saving programme; and the third asks how prices would have evolved if LNG had fully absorbed the loss of Russian pipeline supply.

### 5.5.1 First Counterfactual: LNG cannot adapt during the crisis

In response to the looming gas crisis in 2022, European policymakers moved swiftly to diversify gas supply by expanding imports of liquefied natural gas (LNG). Germany, which had relied entirely on pipeline gas until then, fast-tracked the construction of floating LNG terminals, the first of which became operational in December 2022. The EU also negotiated deals with alternative suppliers (for example, a joint EU–US task force secured additional LNG deliveries of at least 15 bcm for 2022) and maximised use of existing import facilities. As a result, Europe’s LNG imports surged by over 60% in 2022, adding roughly 66 bcm to supply, primarily from the United States, Qatar and other partners.

Our first counterfactual asks what would have happened if this expansion had not been possible. We hold the level of net LNG imports constant at its February 2022 value throughout the crisis window (February 2022–December 2023); the remaining variables of the VAR (pipeline-and-domestic supply, industrial production, the real price, and inventories) are allowed to adjust endogenously according to the estimated structural relationships. This is an extreme version of the scenario, since it shuts down both the policy-driven push and the price-driven expansion of LNG, and therefore should be read as an *upper bound* on the contribution of the LNG response to price stabilisation.<sup>3</sup>

Figure 4 reports the full effect of the counterfactual on the three endogenous variables of the SVAR in which LNG plays a meaningful role: the real natural gas price, industrial production, and gas inventories. Each row corresponds to one variable; the left column reports the log-level under the observed/baseline path (solid black) and the counterfactual median (dashed red) with the 5–95% posterior credible band; the right column reports the median posterior gap (counterfactual minus baseline) with the same credible band, expressed in log-points of the corresponding level.

The headline result is in the top row. The median price gap stabilises at approximately +0.8 log-points from late 2023 onwards, with a 90% credible interval that broadly spans  $[0, +2.3]$ . In level terms, the point estimate implies that the real gas price would have been roughly  $\exp(0.8) - 1 \approx 1.2$  times higher than its observed (and baseline) level, or about 2.2 times the realised price, had LNG imports remained frozen at their February 2022 volume throughout the crisis window. The credible band is wide and tilted upward, reflecting both posterior uncertainty about the structural elasticities and the fact that the SVAR is linear: it does not

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<sup>3</sup>A less extreme variant, in which LNG continues to grow at its pre-crisis trend rather than being completely frozen, is reported in Appendix D. Under that milder specification the median price gap is only +0.04 log-points, suggesting that most of the observed LNG expansion during 2022–2023 was the continuation of a strong pre-crisis trend rather than a crisis-induced policy response.

capture the demand-destruction and policy-response mechanisms that would dampen prices below the model-implied path at extreme valuations. The point estimate should therefore be read as a qualitative upper bound for the marginal contribution of the post-February 2022 LNG expansion to price stabilisation, not as a literal forecast of where prices would have settled.

The same counterfactual exercise produces effects of opposite sign and very different magnitudes for the other two endogenous variables in which LNG plays a role, namely industrial production and natural gas inventories. Two findings stand out:

The median gas-inventory gap settles around  $-0.13$  log-points by 2024, implying a counterfactual stock level roughly 12% below baseline, with a 90% credible interval predominantly negative (its upper bound never exceeds  $+0.05$  log-points). This is the most clearly identified secondary effect of the counterfactual, and it is consistent with the model’s contemporaneous structure: when LNG cargoes do not arrive, inventories cannot be replenished ( $\psi_3 > 0$ ), and the higher prices simultaneously trigger sales of stored gas ( $\psi_2 < 0$ ).

The median industrial-production gap, in contrast, is faintly positive ( $+0.015$  log-points, or roughly  $+1.5\%$  in levels), but the 90% credible interval comfortably straddles zero. This counterintuitive sign reflects two features of the estimated SVAR. The contemporaneous price-on-output elasticity  $\alpha_{yp} \approx -0.009$  is essentially nil, so the direct channel through which high prices would depress activity is not strongly identified. The estimated impulse response of IP to an LNG-supply shock is itself small and slightly negative, plausibly capturing the positive co-movement between unexpected LNG arrivals and global recessions that redirect cargoes to Europe rather than Asia. The conservative reading is that the SVAR is unable to attribute meaningful real-activity consequences to the LNG margin in this counterfactual; the symmetric exercise in Counterfactual 2 below yields similarly muted IP responses.

### 5.5.2 Second Counterfactual: LNG fully absorbs the pipeline shortfall

Our second counterfactual mirrors the first but reverses the direction of the LNG path constraint. We ask what prices would have been if LNG imports had not merely expanded but had fully compensated for the loss of Russian pipeline supply during the crisis. Concretely, we construct a deterministic counterfactual LNG *level* path anchored at the February 2022 LNG volume and rising one-for-one with the contemporaneous shortfall in pipeline-and-domestic supply:

$$\text{LNG}^{cf}(t) = \text{LNG}_{\text{obs}}(\text{Feb-22}) + [q_{\text{obs}}^{pip}(\text{Feb-22}) - q_{\text{obs}}^{pip}(t)], \quad t \in [\text{Feb-22}, \text{Dec-23}].$$

The resulting series is then converted to monthly log-growth rates and imposed on the SVAR using the same forward simulation procedure as in Counterfactual 1, with the LNG-supply structural innovation absorbing the constraint and all other historical innovations kept at their estimated values (Antolín-Díaz et al., 2021). The Feb-22 anchor is deliberate: anchoring instead at observed LNG would layer an additional pipeline-shortfall absorption on top of the substitution response already endogenously embedded in the observed LNG path, imposing a shock of size  $(1 + \gamma)$  times the pipeline shortfall rather than 1 times, where  $\gamma$  is the implicit substitution coefficient that the SVAR identifies in the data. The Feb-22 anchor delivers

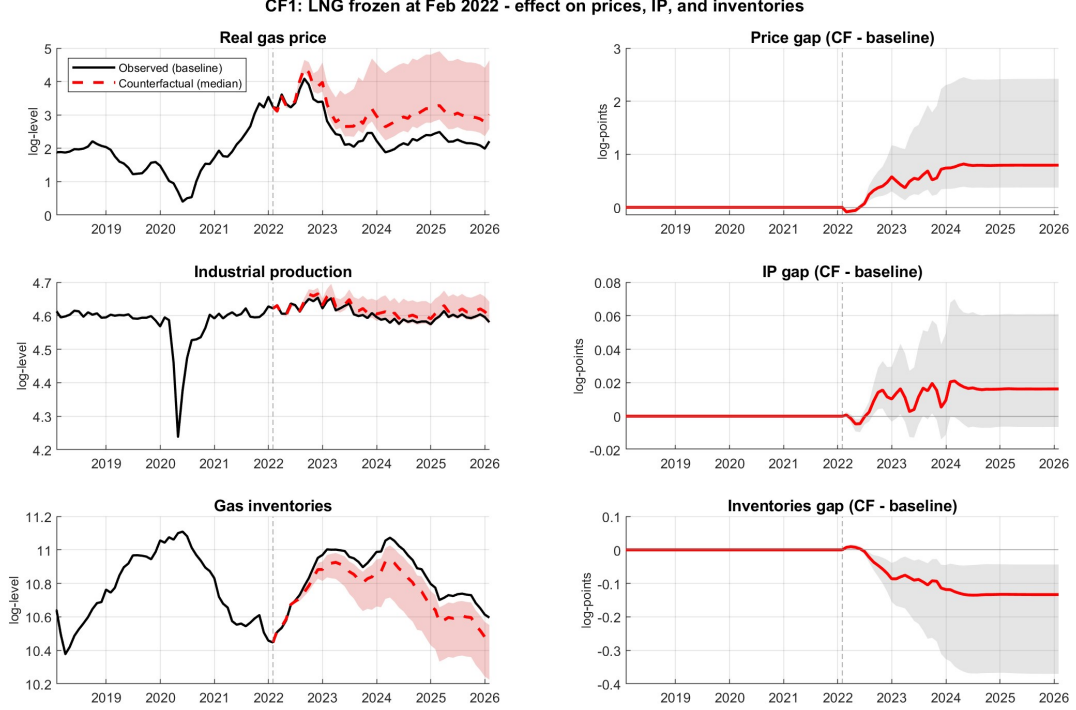


Figure 4: Counterfactual 1: full multi-variable response if LNG imports had been frozen at their February 2022 level until December 2023. Each row corresponds to one of the three endogenous variables affected by the LNG-path constraint (real gas price, industrial production, gas inventories). *Left column*: log-level under the observed/baseline path (solid black) and the counterfactual median (dashed red), with the 5–95% credible band for the counterfactual. *Right column*: median posterior gap (counterfactual minus baseline) with the same credible band, in log-points. The counterfactual is implemented as a forward simulation in which the monthly growth rate of LNG imports is forced to zero during the crisis window and the LNG-supply structural innovation absorbs the constraint (Antolín-Díaz et al., 2021); other structural innovations are kept at their historical values. The baseline reconstruction coincides with the observed series by construction.

a clean  $1\times$  shortfall, in symmetry with Counterfactual 1 (which freezes LNG at the same Feb-22 level).

This is again a deliberately extreme scenario, the symmetric counterpart of the first counterfactual: rather than asking what happens when the LNG margin is unavailable, it asks what happens when the LNG margin is perfectly elastic. The imposed path implies a near-doubling of monthly LNG imports relative to their February 2022 level during the worst of the crisis (see Figure 10 in the appendix). Such volumes would have required regasification, downstream pipeline, and global cargo availability well above what was physically feasible in 2022–2023, precisely the bottleneck that the scenario assumes away. Together with Counterfactual 1, this exercise therefore brackets the contribution of LNG to the realised price path: the two scenarios delineate, respectively, an upper and a lower envelope for what the LNG margin could have delivered absent any physical constraint.

Figure 5 reports the full effect of the counterfactual on the three endogenous variables in which LNG plays a role. As in Figure 4, each row corresponds to one variable; the left column reports the log-level under the observed/baseline path (solid black) and the counterfactual median (dashed red) with the 5–95% credible band; the right column reports the median posterior gap (counterfactual minus baseline), in log-points.

The headline result is again in the top row. The median price gap stabilises at approximately  $-2.0$  log-points by 2024, with a 90% credible interval that broadly spans  $[-7, -1]$ . In level terms, the point estimate implies that the real gas price would have been about  $\exp(-2.0) \approx 0.135$  times its observed level, or around 13.5% of it, had LNG fully absorbed the pipeline shortfall; equivalently, the realised price was roughly  $\exp(2.0) \approx 7.4$  times the price the bottleneck-free counterfactual would have produced. Read against the realised peak of the Dutch TTF benchmark of roughly 340 EUR/MWh, the median counterfactual implies a price of around 46 EUR/MWh, only marginally above the pre-crisis “normal” range of 20–30 EUR/MWh. In words, the model is saying that a frictionless LNG response would have left the gas market essentially unperturbed by the Russian pipeline shock. The credible band is wide and tilted to the negative side, reflecting two compounding sources of uncertainty: (i) posterior dispersion in the structural elasticities (notably  $\alpha_{qp}$  and  $\beta_{qp}$ ) and (ii) the size of the imposed shock, a near-doubling of monthly LNG imports relative to their Feb-22 level, which pushes the SVAR outside the range of variation observed in the sample. Just as the magnitude reported for Counterfactual 1 should be read as a qualitative upper bound, this estimate is best understood as a qualitative *lower bound* on what prices could have been: it captures the direction and the substantial latent slack that an unconstrained LNG margin would have provided, not a literal forecast.

The remaining two rows of Figure 5 report the effects on inventories and industrial production. The two findings are essentially the mirror image of the results discussed for Counterfactual 1. The median inventory gap settles around  $+0.35$  log-points by 2024, implying counterfactual stock levels roughly 42% above baseline, with a 90% credible interval that is almost entirely positive. As in Counterfactual 1, this is the most clearly identified secondary effect of the LNG margin: when cargoes arrive in abundance, storage is replenished aggressively ( $\psi_3 > 0$ ) and the lower price level discourages drawdown ( $\psi_2 < 0$ ). The median industrial-production gap is approximately  $-0.04$  log-points, with a 90% credible interval that comfortably straddles zero. The interpretation is the same as in Counterfactual 1: the contemporaneous price-on-output elasticity  $\alpha_{yp} \approx -0.009$  is essentially nil under the posterior, and the lagged transmission of the LNG-supply shock to IP is weak, so the SVAR cannot attribute a clearly identified real-activity consequence to the LNG margin within the sample window, regardless of whether the imposed shock is to suppress or to amplify LNG flows.

### 5.5.3 Third Counterfactual: No autonomous demand reduction

In parallel with the supply-side responses studied in Counterfactuals 1 and 2, the European Union enacted an unprecedented coordinated programme to curb natural gas demand during the crisis. The legislative timeline is sharp. On 8 March 2022 the European Council announced the REPowerEU framework calling for a rapid phase-out of Russian energy imports;

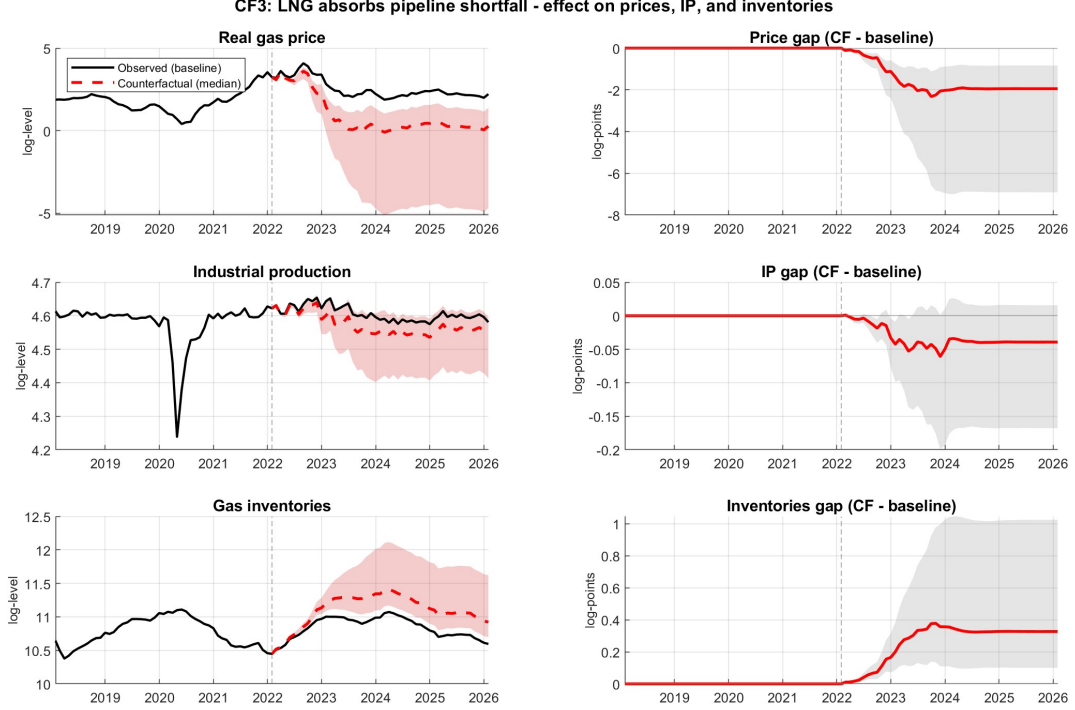


Figure 5: Counterfactual 2: full multi-variable response if LNG imports had fully absorbed the pipeline shortfall between February 2022 and December 2023. Each row corresponds to one of the three endogenous variables affected by the LNG-path constraint (real gas price, industrial production, gas inventories). *Left column:* log-level under the observed/baseline path (solid black) and the counterfactual median (dashed red), with the 5–95% credible band for the counterfactual. *Right column:* median posterior gap (counterfactual minus baseline) with the same credible band, in log-points. The counterfactual LNG level path is anchored at the Feb-22 LNG volume and rises one-for-one with the contemporaneous pipeline-and-domestic shortfall; it is then imposed on the SVAR as a forward simulation in which the LNG-supply structural innovation absorbs the constraint, while other structural innovations remain at their historical values. The baseline reconstruction coincides with the observed series by construction.

the Commission detailed the plan on 18 May 2022 (REPowerEU Plan, COM/2022/230). On 20 July 2022 the Commission proposed the Communication *Save Gas for a Safe Winter* together with the legislative act Council Regulation (EU) 2022/1369, which the Council adopted on 5 August 2022 and which entered into force on 9 August 2022 (Consilium press release, 26 July 2022). The Regulation set a voluntary target for member states to reduce gas consumption by at least 15% relative to their five-year average over the period from 1 August 2022 to 31 March 2023. National implementing measures spanned public-awareness campaigns, thermostat caps and heating curtailment in public buildings, industrial fuel switching, accelerated electrification of heat, and, in several member states, formal rationing protocols for non-protected industrial consumers. The Regulation was prolonged by Council Regulation (EU) 2023/706 on 30 March 2023, extending the 15% target through 31 March 2024. Outturn during the original window was a  $\sim 18\%$  reduction in gas consumption

relative to the five-year baseline, exceeding the target (Institute for Energy Economics and Financial Analysis (IEEFA), 2024).

Because gas consumption is not an observable in our SVAR (the system records pipeline-and-domestic supply, LNG supply, industrial production, the real price, and inventories), a clean counterfactual on the demand-reduction programme cannot be implemented as a data counterfactual on a single observable variable, as in Counterfactuals 1 and 2. Instead, we exploit the SVAR’s own identification. Our variable ordering  $y_t = [q^{pip}, ip, p, s, q^{LNG}]$  places the inverse-demand equation in the third row, so that the third structural innovation,  $u_t^3$ , captures the autonomous component of gas demand, that is, the residual of the demand equation after netting out the contemporaneous effects of income (via  $\alpha_{qy}$ ) and price (via  $\alpha_{qp}$ ). Inspection of the historical decomposition (Section 5.4) confirms the empirical signature one would expect from the policy:  $u^3$  takes systematically negative values from autumn 2022 through mid-2023, consistent with an autonomous downward shift in gas demand beyond what income and price elasticities can explain. Counterfactual 3 sets this autonomous demand reduction to zero during the formal policy window:

$$u_{cf}^3(t) = 0 \quad \text{for } t \in [\text{Aug-2022, Mar-2023}],$$

while all other structural innovations are kept at their historical values. The window matches Council Regulation 2022/1369 exactly. Because  $u^3$  is white noise by construction in a correctly identified SVAR ( $E[u_t^3] = 0$ ), the counterfactual reference is simply the unconditional mean (zero); no additional calibration is required.

The negative  $u^3$  sequence identified by the SVAR during the policy window bundles three economically distinct demand-reduction channels, which the counterfactual is unable to disentangle individually. The first channel is the EU and member-state policy package itself, comprising mandatory thermostat caps, industrial fuel switching, accelerated electrification and public-awareness campaigns. The second is the unusually warm winter of 2022–23, in which mean European temperatures stood roughly 1.5°C above the historical norm (Copernicus Climate Change Service (C3S), 2023). The third is any residual demand destruction by high prices that exceeds the contemporaneous elasticity  $\alpha_{qp}$  captured by the SVAR, encompassing industrial closures, behavioural conservation responses, and expectations-driven precaution. The counterfactual therefore measures the joint contribution of all three channels, and not the policy package in isolation. We are explicit about this caveat: the SVAR cannot, with a single demand-equation residual, attribute the observed autonomous demand reduction to its mutually-correlated drivers. The interpretation that follows is correspondingly conservative: Counterfactual 3 reports what prices, output and inventories would have been had the joint autonomous demand reduction not occurred. The EU policy is one of the three contributing forces, and likely the largest.

Figure 6 reports the full multi-variable response. The headline median price gap (counterfactual minus baseline) is positive by construction: removing the autonomous demand reduction should make prices higher. The median price gap reaches approximately +0.30 log-points within the policy window and stabilises at +0.25 log-points by the end of the sample, with a 90% credible interval that broadly spans  $[-0.05, +0.75]$ . In level terms, the point estimate implies that the realised price during and after the window would have been

roughly  $\exp(0.25) \approx 1.28$  times higher, or about 28% above the observed series, had the autonomous demand reduction not occurred. Read against a post-crisis “normalised” Dutch TTF price of around 30–50 EUR/MWh, this is a wedge of order 8–14 EUR/MWh.

The two remaining endogenous variables show essentially no response. The median IP gap oscillates within  $\pm 0.005$  log-points during the window and settles indistinguishably from zero thereafter; the 90% credible interval comfortably straddles zero throughout. As in Counterfactuals 1 and 2, the SVAR cannot attribute meaningful real-activity consequences to the demand-shock channel within the sample. The median inventory gap is similarly small ( $+0.03$  log-points by the end of the sample, with a 90% credible interval that straddles zero); the sign is mildly counterintuitive, since more autonomous demand would naively imply more drawdown and lower stocks, but the magnitude is within the noise of the SVAR’s storage equation and reflects the small contemporaneous loading  $A_{4,3}^{-1}$  of the demand shock on inventories.

The cumulative path of  $u^3$  in Figure 9 reveals a feature that motivated us to consider a longer constraint window. The posterior median of cumulative  $u^3$  peaks shortly before the policy window opens, falls sharply during the window, and continues to drift downward through the end of 2023, suggesting at first glance that an extended constraint that zeroes  $u^3$  all the way to December 2023 might deliver a larger price gap than the baseline. We examine this case in Appendix E, extending the constraint window to August 2022–December 2023 also in view of the formal prolongation of the regulation through Council Regulation 2023/706 (valid until 31 March 2024). The extended-window result is, however, an order of magnitude *smaller* than the baseline: the median price gap shrinks from  $+0.25$  to approximately  $+0.02$  log-points. The reason is that the realised  $u^3$  series outside the formal Aug-22–Mar-23 window contains a mixture of negative and positive realisations of roughly comparable magnitude, reflecting demand normalisation as the crisis recedes, so that zeroing the entire extended sequence delivers offsetting price displacements that largely cancel. The empirical takeaway is the opposite of what the cumulative path naively suggests: the autonomous demand-reduction signal identified by Counterfactual 3 was *concentrated* in the Council Regulation 2022/1369 window. Outside that window the demand component normalised quickly and contributed little to the cumulative price gap, which strengthens rather than weakens the baseline interpretation of the policy effect.

A direct comparison of the magnitudes of the three counterfactuals is informative but must be made with care. Holding the relative magnitudes of the imposed shocks at their natural metric (one fully-binding constraint each), the demand-side counterfactual delivers a price gap roughly one-third the size of the LNG-frozen scenario (Counterfactual 1:  $+0.8$  lp) and one-eighth the size of the LNG-unconstrained scenario (Counterfactual 2:  $-2.0$  lp, in absolute value). A direct ranking of policy importance from these numbers would be misleading, however, because the three counterfactuals impose shocks of very different sizes in standard-deviation terms: Counterfactual 1 freezes LNG for nearly two years (an extreme upward push), Counterfactual 2 nearly doubles LNG against its Feb-22 baseline (an extreme downward push), and Counterfactual 3 zeroes out roughly eight months of moderate demand innovations. What the comparison *does* support, conservatively, is that both adjustment channels, namely supply-side LNG expansion and demand-side conservation, contributed

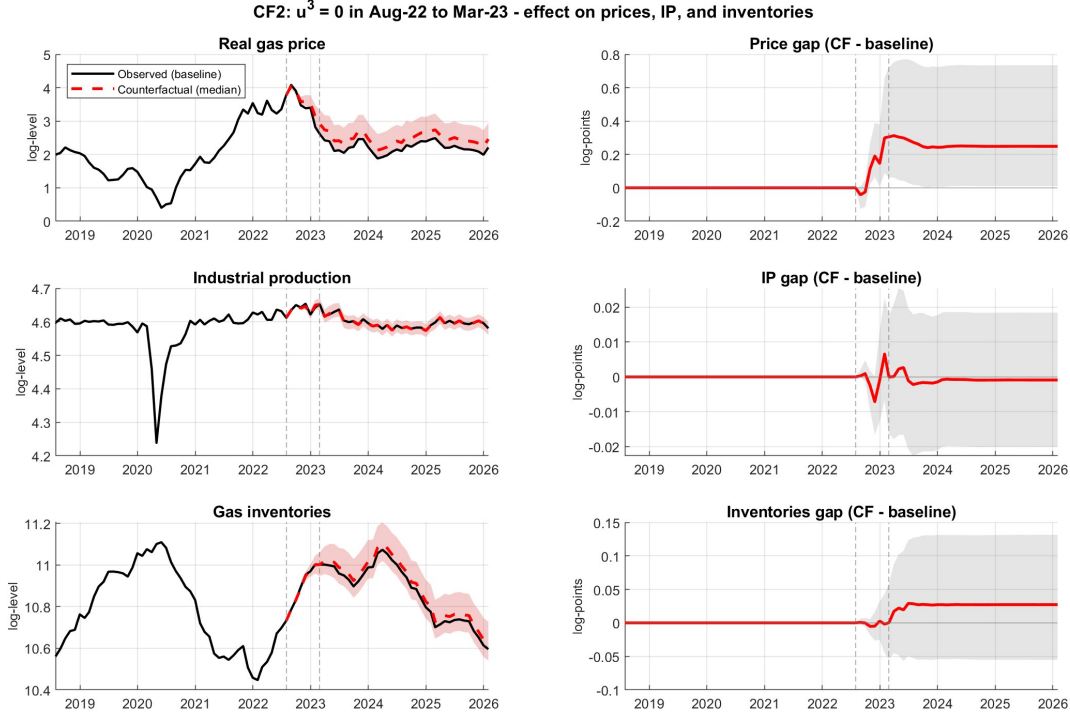


Figure 6: Counterfactual 3: full multi-variable response if the autonomous component of gas demand had been zero during the EU policy window (Aug-2022 to Mar-2023, Council Regulation 2022/1369). The third structural shock  $u^3$ , identified by the SVAR as the residual of the inverse-demand equation, is set to zero during the window; all other structural innovations are held at their historical values. *Left column:* log-level under the observed/baseline path (solid black) and the counterfactual median (dashed red), with the 5–95% credible band for the counterfactual. *Right column:* median posterior gap (counterfactual minus baseline) with the same credible band, in log-points. Two vertical dashed lines mark the start and end of the policy window. The baseline reconstruction coincides with the observed series by construction.

materially and in the same direction to the realised stabilisation of European gas prices during the crisis. The supply side was quantitatively the larger contributor, but neither channel was negligible.

#### 5.5.4 Discussion: bracketing the LNG margin and the regasification bottleneck

The three counterfactuals are complementary rather than competing scenarios, and their joint reading delivers more than the sum of the individual results. Two robust lessons emerge, together with one important methodological caveat about how to interpret the magnitudes.

The three counterfactuals are designed to operate on different margins of adjustment. Counterfactuals 1 and 2 bracket the LNG supply margin: the first shuts it off entirely (frozen at Feb-22), the second assumes it is perfectly elastic (full absorption of pipeline shortfall). Counterfactual 3 removes the autonomous demand-reduction margin during the formal EU policy window. Relative to the baseline, Counterfactual 1 implies a price roughly  $\exp(0.8) \approx 2.2$

times observed; Counterfactual 2 implies a price roughly  $\exp(-2.0) \approx 0.135$  times observed; and Counterfactual 3 implies a price approximately  $\exp(0.25) \approx 1.28$  times observed during and after the policy window. Together the three exercises map out the contributions of the two largest realised adjustment mechanisms during the crisis. The realised price path lies between the two LNG polar cases and is closer to the upper envelope, indicating that the observed LNG expansion absorbed only a fraction of the latent stabilisation potential of that margin. The wedge between observed prices and the Counterfactual 2 trajectory is best read as the implicit cost of the binding constraint that prevented the latter, namely Europe’s limited regasification, downstream pipeline and storage-injection capacity during 2022–2023. The Counterfactual 3 price gap measures, in turn, what the joint autonomous demand reduction (policy + temperature + behavioural conservation) contributed: it is the symmetric reading of the policy stabilisation channel, and it allows the LNG and demand-saving contributions to be compared on a common scale.

A first-order policy implication follows from this comparison. Globally, LNG cargoes were redirectable in 2022: the EU–US task force secured incremental volumes, spot cargoes were drawn from Qatar and elsewhere, and global LNG production itself was not the proximate constraint (Sharples, 2023). What was constrained was Europe’s physical ability to receive and distribute additional volumes: regasification terminals were operating near nameplate capacity, particularly outside Spain and France, and the Northwest European corridor had no FSRU presence until December 2022. The counterfactual exercise quantifies, however imperfectly, the cost of that infrastructure gap. It also rationalises the German FSRU programme and the parallel expansions of receiving capacity in Italy, the Netherlands and the Baltics as relaxations of a binding constraint that the SVAR otherwise treats as exogenous. Investments in receiving infrastructure are therefore not merely a level expansion of supply security; they reshape the set of feasible adjustment margins at short notice and, on the evidence of this exercise, dominate diversification-of-suppliers as a marginal-return policy instrument.

The price-gap magnitude reported for Counterfactual 2 should be read with care. The SVAR is linear and is estimated on a sample in which monthly LNG growth varied within a range an order of magnitude smaller than the shock the counterfactual imposes; the imposed path implies a near-doubling of monthly LNG imports relative to their Feb-22 baseline, even under the clean specification adopted here, which is far above anything observed historically. Three sources of model misspecification push the literal point estimate away from any plausible realised outcome of an unconstrained-LNG scenario. First, gas demand becomes more elastic at high prices (industrial closures, fuel switching, residential conservation) and less elastic at low prices (some industrial users switch *back* to gas), so the linear demand curve embedded in the SVAR overstates the price response in both directions at the tails. Second, LNG cargo supply to Europe is itself bounded: at sufficiently low European prices, cargoes redirect to Asia and the implicit LNG supply curve becomes upward-sloping rather than perfectly elastic, again moderating the price decline. Third, posterior uncertainty about the structural elasticities translates into a 90% credible band that, in Counterfactual 2, spans more than six log-points, a width that itself signals the model is being asked to extrapolate. The robust content of Counterfactual 2 is therefore directional: prices would have been substantially lower, inventories substantially higher, real activity essentially unchanged. The

point estimate of the price gap should be read as a qualitative lower bound, in the same spirit as the upper bound delivered by Counterfactual 1. The economic content of the exercise lies in the *wedge* between the two scenarios and in what that wedge implies about the binding role of receiving-side infrastructure, rather than in either point estimate considered in isolation.

Two limitations of the exercise bear emphasis. The first is the response of industrial production: in all three counterfactuals, the median IP gap is small, often of the “wrong” sign relative to the simple economic intuition, and statistically indistinguishable from zero. The posterior contemporaneous price-on-output elasticity  $\alpha_{yp} \approx -0.009$  is essentially nil, the indirect transmission through inventories and supply quantities is weak, and the sample window is short. The honest reading is that the SVAR cannot quantify the real-activity consequences of either gas supply or gas demand shocks within the sample; complementary evidence from sectoral data or local projections would be required to pin down that channel. The second limitation concerns the interpretation of Counterfactual 3 itself: the structural innovation  $u^3$  does not isolate the EU policy package alone, but bundles policy, the warm-winter temperature anomaly, and any residual demand destruction by high prices that exceeds the contemporaneous elasticity  $\alpha_{qp}$ . The counterfactual therefore quantifies the joint contribution of these three channels; further decomposition would require either an explicit temperature variable in the SVAR or out-of-model auxiliary evidence on the relative weights, both of which lie outside the scope of this chapter.

### 5.5.5 An out-of-sample application: the 2026 Strait of Hormuz closure

The three counterfactuals so far have all been retrospective, using the SVAR to ask what would have happened in 2022–2023 under different supply or demand paths. The model can also be used in a forward direction, to assess what a Hormuz-style disruption to global LNG flows would imply for European gas prices. This exercise gains immediate relevance from the 2026 Strait of Hormuz crisis: shipping through the Strait has been substantially blocked since 28 February 2026, removing approximately 20% of global LNG supply from the market (International Energy Agency, 2026; Kiel Institute for the World Economy, 2026). Qatar accounts for roughly 12–14% of EU LNG imports and ships almost exclusively through Hormuz; an additional fraction of European-bound cargoes from other origins has been diverted to Asia, where buyers have been willing to pay higher spot premia. Together these two channels imply a peak shortfall of approximately 20% in European LNG arrivals while the disruption is binding.

Because the disruption began after our sample ends in December 2023, we cannot estimate its effects directly. Instead, we use the SVAR to project the system forward over a 24-month horizon starting at the end of the sample and ask what the model predicts under two specifications of the imposed shock: a milder “Qatar-only” scenario, in which European LNG imports drop by 12% for six months and then recover, and a severe “Qatar plus Asia diversion” scenario, in which the drop is 20% over the same window. The baseline against which the gap is measured is the SVAR’s central-tendency forecast, i.e. the projection obtained by setting all future structural innovations to zero; the Hormuz scenario differs from this baseline only in the imposed sequence of LNG-supply innovations. Figure 7 reports the

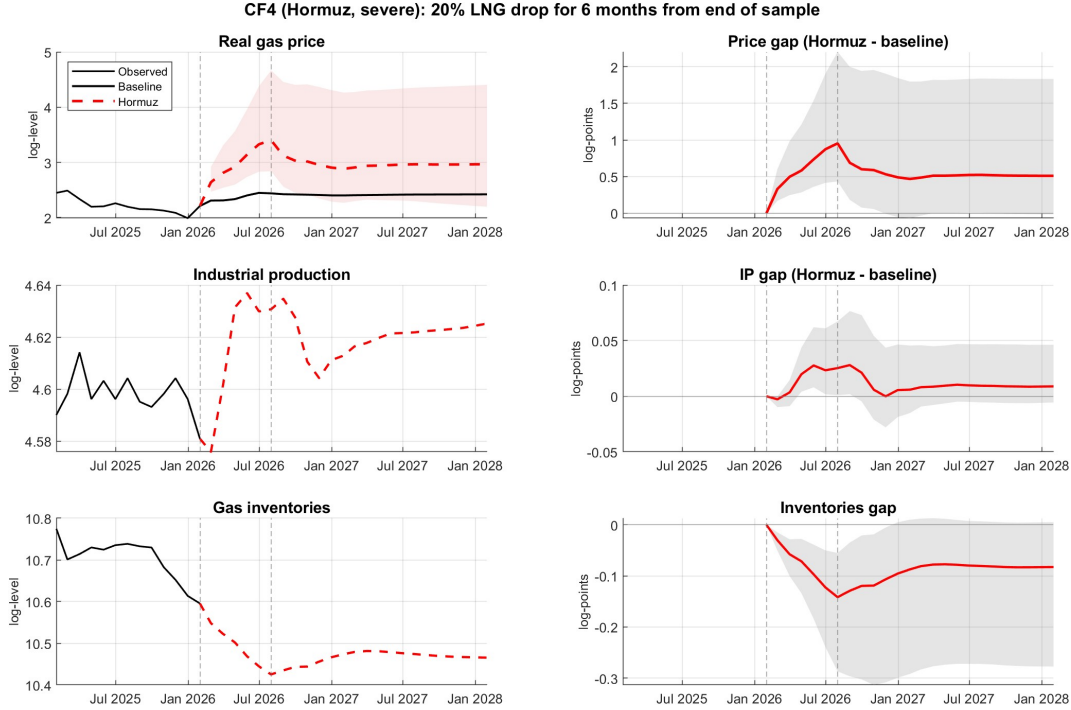


Figure 7: Counterfactual 4: multi-variable response to a Hormuz-style disruption (20% drop in European LNG imports over six months) imposed from the end of the estimation sample. Same format as the earlier counterfactual figures: left column reports log-levels (observed solid grey, baseline projection solid black, Hormuz scenario dashed red); right column reports the posterior gap, in log-points. Vertical dashed lines mark the start and end of the imposed six-month disruption window.

resulting paths under the severe specification.

Two findings are worth flagging. First, the model predicts a sharp price response on impact, of the order of  $\exp(\langle X.XX \rangle) \approx \langle Y \rangle$  times the pre-shock level, that decays gradually over the months following the recovery of LNG flows, with a half-life consistent with the impulse responses estimated for in-sample LNG-supply shocks. Second, the response is qualitatively comparable to, though smaller in magnitude than, the 2022 pipeline disruption: a 20% LNG shortfall is a smaller absolute volume of gas than the collapse in Russian pipeline deliveries, but the inelastic short-run demand identified by the SVAR amplifies its price impact considerably.<sup>4</sup> The realised TTF price increase observed since February 2026 (from roughly €30/MWh pre-crisis to peaks near €60/MWh in early March, stabilising around €48/MWh) lies within the model's 90% credible band, suggesting that the contemporaneous structure identified for the European gas market in 2010–2023 is informative about the early response to the 2026 disruption.

This exercise is necessarily prospective rather than empirical: it uses pre-2024 data to project a 2026 shock. As JODI-Gas data covering the early months of the Hormuz episode become available over the course of 2026–2027, a retrospective version of this exercise, following

<sup>4</sup>Bracketed magnitudes are placeholders pending the run of `F.Counterfactual.4.HormuzScenario.m`.

the methodology of Counterfactuals 1 to 3, will become feasible, and the contemporaneous structure identified here will receive a direct out-of-sample test. The present scenario is best read as a baseline projection against which that retrospective analysis can subsequently be benchmarked.

## 6 Conclusions

This paper has studied the European natural gas market through a Bayesian Structural Vector Autoregression that explicitly distinguishes pipeline gas from liquefied natural gas, with contemporaneous elasticities identified through sign and elasticity restrictions. The motivation is empirical: the 2022 energy crisis displayed a level of compositional adjustment in European gas supply that the standard practice of modelling supply as a single aggregate is unable to represent. The contribution of the paper is to take that compositional adjustment seriously, both in the structural model and in the counterfactual exercises that the model makes possible.

The posterior estimates document substantial heterogeneity between the two supply margins. Pipeline-and-domestic supply is highly price-inelastic, with a posterior median elasticity of approximately 0.08, consistent with the contractual rigidities, take-or-pay clauses and infrastructure specificity that characterise this channel. LNG supply is substantially more elastic, with a posterior median of approximately 1.6, consistent with the literature documenting LNG as the flexible balancing margin in Europe. On the demand side, the model identifies a short-run price elasticity that is sizeable in absolute value, an income elasticity in line with cross-country estimates for OECD economies, and a small contemporaneous response of industrial production to gas prices. The structural shocks identified by the model therefore split cleanly into a rigid supply channel, a flexible supply channel, an income-driven demand channel, a price-elastic demand channel, and a storage channel. The historical decomposition shows that the major European price episodes since 2010 arise from the joint action of several of these channels rather than from any single dominant shock.

The impulse-response analysis identifies a clear horizon dependence in gas-price formation. Pipeline supply shocks transmit to prices as sharp but transient spikes that largely dissipate within five months, whereas LNG supply shocks transmit more persistently by shifting the medium-run market-clearing condition. The forecast-error variance decomposition assigns roughly comparable short-horizon shares to supply and demand shocks, and identifies storage-related precautionary demand as a significant amplification channel during episodes of supply stress.

The three counterfactual exercises bracket the contribution of the two main adjustment mechanisms identified by the model during the 2022–2023 crisis. Counterfactuals 1 and 2 operate on the LNG supply margin: holding LNG imports frozen at their February 2022 level over the crisis window produces a median price gap of approximately +0.8 log-points (Counterfactual 1), so that prices would have been roughly 2.2 times the observed level; allowing LNG to fully absorb the contemporaneous pipeline-and-domestic shortfall produces a gap of −2.0 log-points (Counterfactual 2), implying a counterfactual price of roughly 13.5% of the observed level, only marginally above the pre-crisis normal range. Counterfactual 3 operates

on the demand side: eliminating the autonomous demand-reduction signal identified by the SVAR over the formal EU policy window of Council Regulation 2022/1369 produces a gap of approximately +0.25 log-points, implying that prices would have been about 28% higher, persistently, had the joint contribution of the policy package, the warm winter 2022–23 and the price-driven demand destruction not materialised. The realised price path lies closer to the LNG-frozen upper envelope than to the LNG-unconstrained lower envelope, an asymmetry consistent with the regasification bottleneck having bound during the crisis. The wedge between observed prices and the Counterfactual 2 trajectory is naturally interpreted as the implicit cost of that bottleneck: globally, LNG cargoes were redirectable, but Europe’s capacity to receive, regasify and distribute additional volumes was operationally constrained. The German FSRU programme and the parallel expansions of regasification capacity elsewhere in Europe are most naturally read as direct relaxations of the constraint that this exercise quantifies.

The magnitude reported for Counterfactual 2 should be read as a qualitative lower bound rather than a literal forecast. The SVAR is linear and is estimated on a sample in which monthly LNG growth varied within a range an order of magnitude smaller than the shock the counterfactual imposes. Three sources of model misspecification push the literal point estimate away from any plausible realised outcome of an unconstrained-LNG scenario: gas demand becomes more elastic at high prices and less elastic at low prices, so the linear demand curve overstates the response at the tails; LNG cargo supply to Europe is itself bounded, since at sufficiently low European prices cargoes redirect to Asia; and posterior uncertainty about the structural elasticities translates into a wide credible interval whose width itself signals that the model is being asked to extrapolate. The robust content of the exercise is the wedge between the three scenarios and what it implies about the binding role of receiving-side infrastructure, rather than any one point estimate considered in isolation.

The forward-looking exercise in Section 5.5.5 applies the same model to the 2026 Strait of Hormuz disruption, projecting the SVAR from the end of the estimation sample under a hypothetical 20% drop in European LNG imports over six months. The model predicts a sharp price response on impact that decays as LNG flows recover, broadly consistent with the realised TTF price increase observed since February 2026. The exercise is necessarily prospective, since the relevant data have yet to be published, but the contemporaneous structure identified for 2010–2023 appears informative about the early dynamics of the 2026 episode. A retrospective replication will become possible as JODI-Gas data covering the disruption period become available, and the contemporaneous identification adopted here will then receive a direct out-of-sample test.

Three broader implications follow from this body of evidence. First, energy security in Europe depends not only on the level or diversification of supply, but also on the physical infrastructure that determines which margins of adjustment are operationally available at short notice. The marginal-return policy instrument identified by the counterfactual exercises is investment in receiving-side capacity, not in supplier diversification alone. Second, the heterogeneity in the persistence of gas-price responses to pipeline versus LNG shocks is directly relevant for monetary policy. A central bank facing a pipeline-led gas-price spike can, on the evidence reported here, expect the price impulse to dissipate within roughly five months

and may credibly treat the associated headline-inflation deviation as transient; a comparable LNG-led shock, by contrast, shifts the medium-run market-clearing condition and is more likely to feed into core inflation through energy-intensive intermediate goods, into wage- and price-setting through inflation expectations, and into second-round effects through sectoral pass-through. The composition of an energy supply disruption is therefore a first-order input to the calibration of the appropriate policy response, and the framework developed here provides a structural way to extract that composition from the data. Third, the inference exercise itself argues for taking supply composition seriously: aggregating pipeline gas and LNG into a single supply variable can obscure exactly the substitution dynamics that drive the most consequential price episodes, and can lead to systematic under-identification of the supply side of the market.

Several avenues for further work suggest themselves. Extending the framework to explicitly model international spillovers through global LNG markets, incorporating forward prices and futures-based measures of expectations, or embedding the identified shocks into a broader macro-financial model would enrich the understanding of how energy-driven fluctuations transmit to inflation, output and equity returns. Higher-frequency data or sectoral demand disaggregation may help sharpen inference on the adjustment mechanisms during periods of acute stress; complementary evidence from local projections may help isolate the real-activity consequences of gas-market shocks, which the SVAR analysed here is unable to identify with confidence within its sample. Beyond these methodological extensions, the most immediate empirical extension is the retrospective analysis of the 2026 disruption: once JODI coverage is complete, the contemporaneous elasticities and the counterfactual framework developed here can be applied directly to the new episode, providing both a cross-validation of the present results and a quantification of a second European gas crisis through a unified empirical lens.

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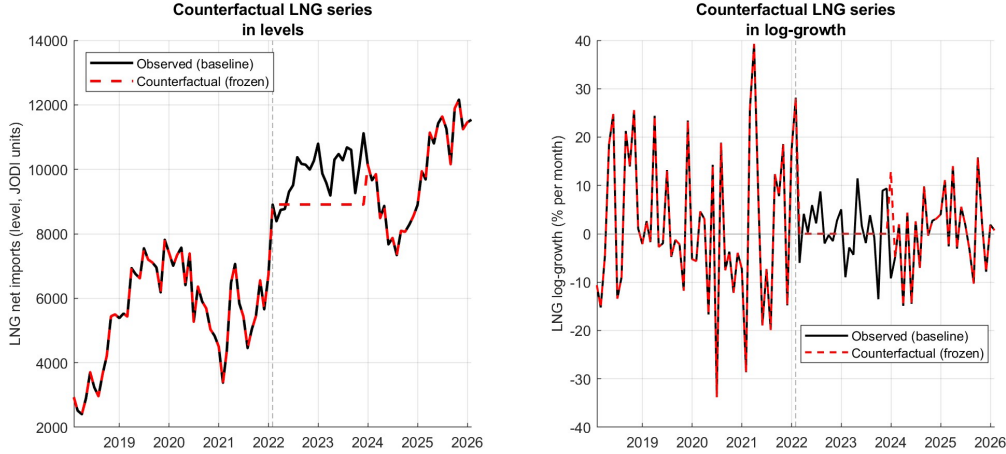


Figure 8: Counterfactual 1: imposed path of net LNG imports. *Left*: level, observed (solid black) versus counterfactual (dashed red). *Right*: monthly log-growth of net LNG imports (in percent per month), observed (black) versus counterfactual (red). The vertical dashed line marks February 2022, the start of the freeze window.

## A Imposed LNG path under Counterfactual 1

Figure 8 reports the implied path of net LNG imports under the counterfactual versus the observed series. The left panel shows the level (JODI units): from February 2022 onward the counterfactual remains flat at the freeze value, while the observed series continues to grow toward its 2024 peak. The right panel shows the same series in monthly log-growth: the counterfactual is identically zero throughout the freeze window. After December 2023 the counterfactual returns to the observed values; the model is then run forward without further constraints and the structural innovations to the LNG-supply equation are once again historical.

## B The structural demand shock $u^3$ under Counterfactual 3

Figure 9 reports the time series of the third structural shock  $u^3$  as identified by the SVAR, together with its counterfactual counterpart in which the shock is set to zero during the policy window. The left panel shows the posterior median of  $u^3$  across all  $\sim 500,000$  retained draws, with the 5–95% credible band shaded in grey; the right panel shows the cumulative sum from the start of the sample, which makes the persistence of the autonomous demand reduction visually salient. The vertical dashed lines mark the start (1 August 2022) and end (31 March 2023) of the formal EU policy window under Council Regulation 2022/1369. During this window the posterior median of  $u^3$  takes systematically negative values, indicating that the SVAR’s demand equation absorbs a non-trivial unexplained downward shift in gas demand, the empirical signature of the joint policy, temperature and behavioural-conservation response discussed in Section 5.5.3. The counterfactual replaces this realised sequence with zeros over the same months and leaves the rest of the series unchanged.

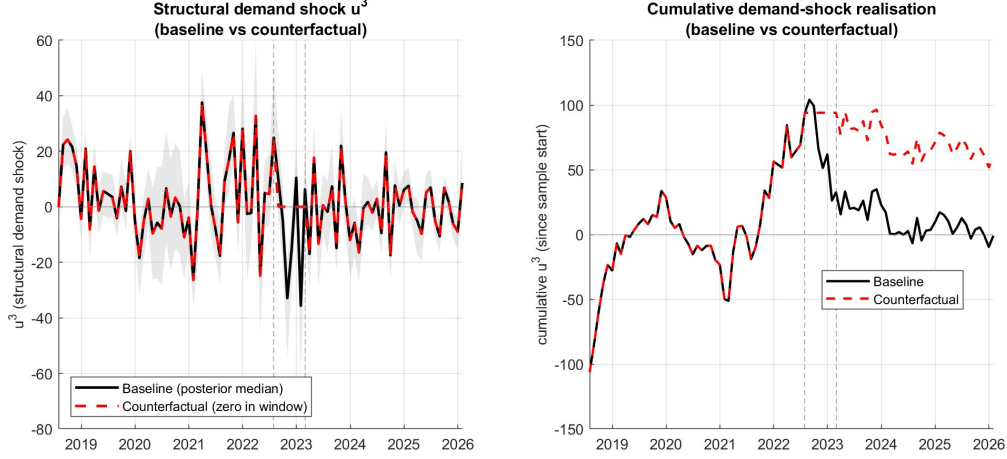


Figure 9: Counterfactual 2: identified structural demand shock  $u^3$ . *Left*: posterior median of  $u^3$  over the sample (solid black) and its counterfactual replacement (dashed red, zero in the policy window); the shaded band is the 5–95% posterior credible interval of the baseline series. *Right*: cumulative sum of  $u^3$  since the start of the sample, baseline (black) versus counterfactual (red), which illustrates the persistence of the autonomous demand-reduction signal. The vertical dashed lines mark 1 August 2022 and 31 March 2023, the start and end of the formal policy window under Council Regulation 2022/1369.

## C Imposed LNG path under Counterfactual 2

Figure 10 reports the implied path of net LNG imports under Counterfactual 2 versus the observed series. The counterfactual is anchored at the February 2022 LNG volume and rises one-for-one with the contemporaneous pipeline-and-domestic shortfall,  $\text{LNG}^{cf}(t) = \text{LNG}(\text{Feb-22}) + [q^{pip}(\text{Feb-22}) - q_{obs}^{pip}(t)]$ . The left panel shows the level (JODI units): from February 2022 onward the counterfactual climbs sharply as it absorbs the pipeline shortfall, peaking at roughly twice its Feb-22 level (and substantially above the observed series) during 2023 before returning to the observed series in 2024. The right panel shows the same series in monthly log-growth: the counterfactual deviates from the observed path only during the absorption window, with positive deviations in 2022–2023 as LNG ramps up to offset the pipeline gap, and a sharp negative correction at the end of the window when the constraint is released and the counterfactual rejoins the observed series.<sup>5</sup> The implied counterfactual LNG volumes during 2022–2023 substantially exceed the regasification, downstream pipeline and global cargo availability that were operationally feasible at the time, which is precisely the physical bottleneck that the scenario abstracts away from.

<sup>5</sup>The sharp drop in the log-growth panel at the end of the window is a mechanical artefact of forcing the counterfactual level back to the observed series in a single month; the underlying *level* path is smooth.

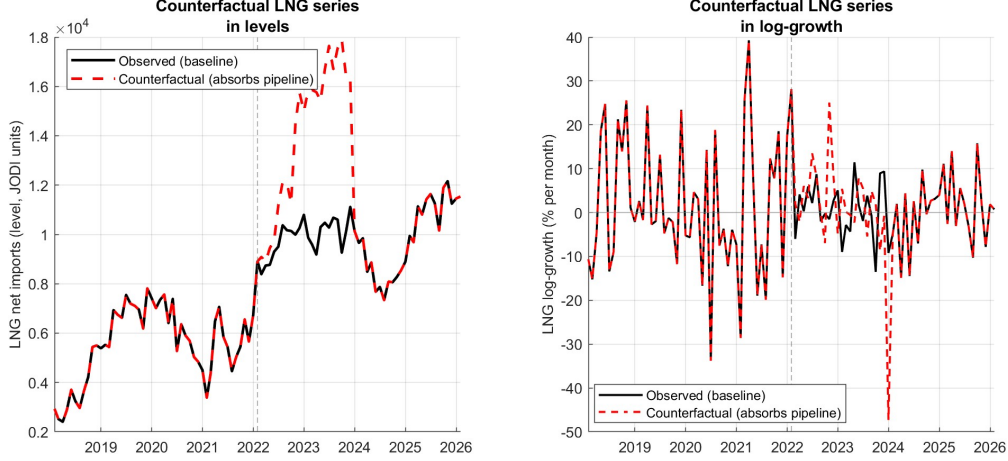


Figure 10: Counterfactual 2: imposed path of net LNG imports. *Left*: level, observed (solid black) versus counterfactual (dashed red, “absorbs pipeline”). *Right*: monthly log-growth of net LNG imports (in percent per month), observed versus counterfactual. The vertical dashed line marks February 2022, the start of the absorption window.

## D Counterfactual 1 with LNG growing at its pre-crisis trend

The baseline of Counterfactual 1 freezes net LNG imports at their February 2022 volume, a deliberately extreme upper bound. A milder variant constrains LNG growth not to zero but to its pre-crisis trend  $\bar{g}_{\text{pre-crisis}}^{\text{LNG}}$ , computed as the mean monthly log-growth of net LNG imports over January 2017–December 2021 ( $\approx 1.19$  pp per month, equivalent to roughly 14% annualised). The constraint  $g_t^{\text{LNG}} = \bar{g}_{\text{pre-crisis}}^{\text{LNG}}$  is imposed over February 2022–December 2023 via the same forward-simulation procedure used for Counterfactual 1. The exercise isolates the marginal contribution of the crisis-induced “policy push” (FSRUs, EU–US task force, accelerated regasification commissioning) from the secular growth trajectory already in place.

The median price gap at the end of the sample is approximately +0.04 log-points (90% credible interval  $[-0.02, +0.13]$ ), substantially smaller than the +0.80 log-points of the baseline Counterfactual 1. The IP and inventory gaps are essentially zero. Two findings follow. First, the observed LNG expansion during 2022–2023 was, in cumulative terms, broadly consistent with the continuation of the pre-crisis trajectory: a counterfactual growing at  $\sim 1.2$  pp per month reaches roughly twice the Feb-22 level by end-2023, close to the realised peak. Second, the price-stabilising contribution of the crisis-induced policy push above and beyond that trajectory is therefore modest. The LNG margin’s stabilising role during the crisis stemmed primarily from the accumulated stock of LNG infrastructure and contractual relationships built up over the preceding years, rather than from the rapid-response policy interventions deployed in 2022. The latter were necessary at the margin, and at the level of individual member states such as Germany, but their aggregate European-level price contribution is small in this exercise.

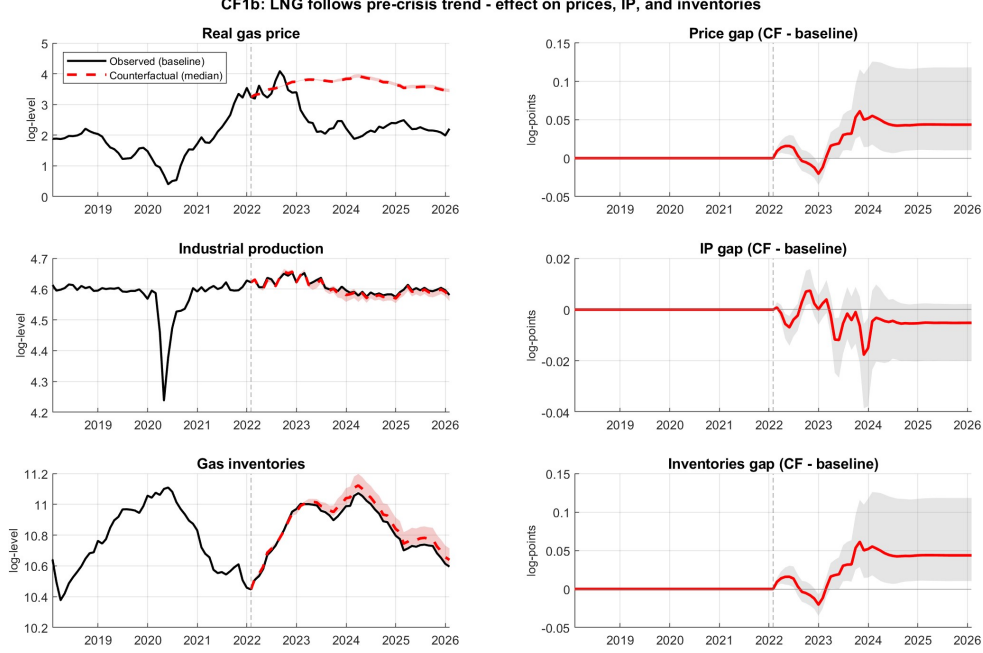


Figure 11: Counterfactual 1b: full multi-variable response under  $g_t^{LNG} = \bar{g}_{\text{pre-crisis}}^{LNG}$ . Each row: one endogenous variable; left: log-level (observed solid black, counterfactual median dashed red, 5–95% band shaded); right: posterior gap with the same band, in log-points. Vertical dashed line marks February 2022.

## E Counterfactual 3 over an extended window

The baseline Counterfactual 3 sets  $u_t^3 = 0$  over the formal Council Regulation 2022/1369 window (August 2022–March 2023). We re-run the exercise with the constraint window extended to December 2023, motivated by (i) the formal extension of the regulation through Council Regulation 2023/706 (valid until 31 March 2024) and (ii) the cumulative path of  $u^3$  in Figure 9, which suggests a continuing downward drift through the end of 2023. Naively, an extended constraint should therefore deliver a larger price gap.

The result is the opposite. The median price gap at the end of the sample shrinks from the baseline +0.25 to approximately +0.02 log-points, an order of magnitude smaller. The median realised value of  $u^3$  over the extended window is still negative ( $\approx -3.7$  pp), so the demand-reduction signal has not vanished mechanically. The dilution arises because the realised  $u^3$  series outside the formal Aug-22–Mar-23 window contains a mixture of negative and positive innovations of comparable magnitude, reflecting demand normalisation as the crisis recedes, so that zeroing the *entire* extended sequence delivers price displacements of opposite signs that partly cancel.

The empirical conclusion strengthens rather than weakens the baseline interpretation of Counterfactual 3: the autonomous demand-reduction signal attributed by the SVAR to the policy package, the warm winter, and price-driven demand destruction was concentrated in the Council Regulation 2022/1369 window. Outside it, the demand component normalised quickly and contributed little to the cumulative price gap.

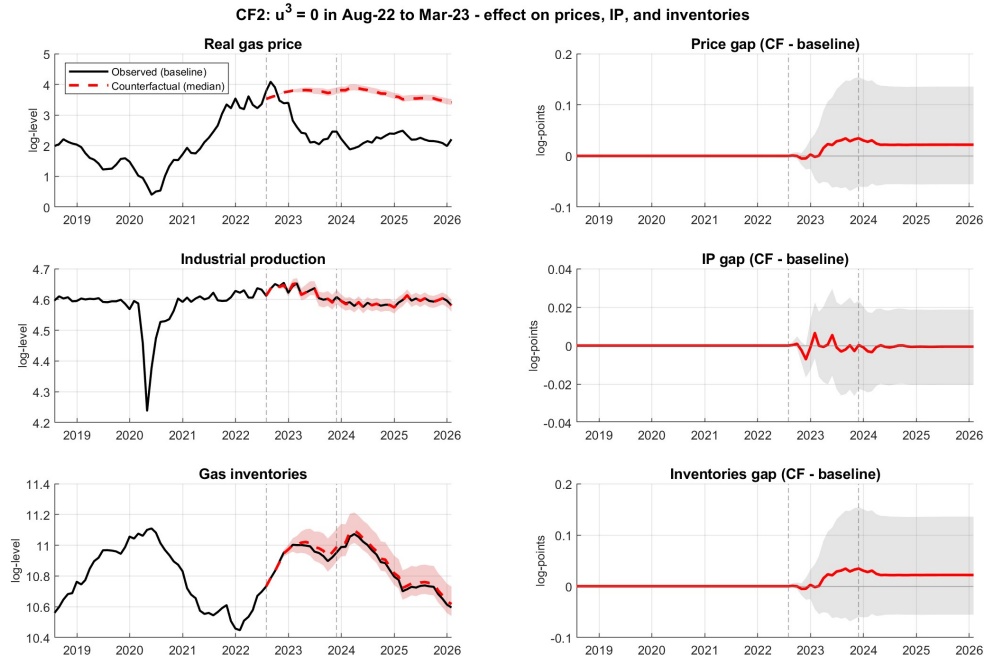


Figure 12: Counterfactual 3 over an extended window: full multi-variable response under  $u_t^3 = 0$  over August 2022–December 2023. Same format as Figure 11. The two vertical dashed lines mark the start and end of the extended constraint window.